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Questions

Q1.

This question concerns six isomers each with the molecular formula $C_5H_{10}O_2$.

(a) Isomers **A**, **B**, **C** and **D** are structural isomers that all react with aqueous sodium carbonate to produce carbon dioxide.

Isomer **A** is a straight-chain compound.
Isomer **B** has only two peaks in its high resolution proton NMR spectrum, both of which are singlets.
Isomer **C** contains a chiral centre but isomer **D** does not.
Give the structures of these **four** isomers.

(4)

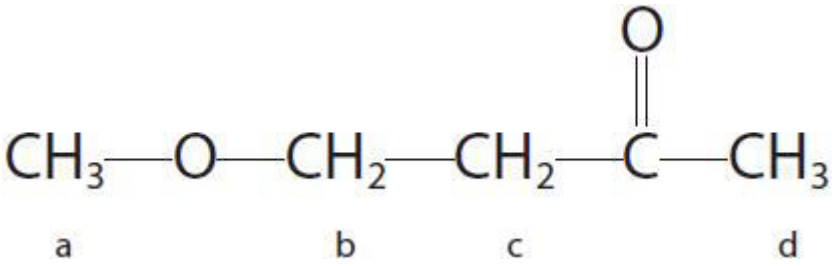
A	B

C	D

(b) (i) Isomer **E** contains an ether functional group (R-O-R) and a ketone functional group. The **low** resolution proton NMR spectrum of isomer **E** has four peaks. The displayed formula of isomer **E** is

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The four hydrogen environments responsible for the four peaks are labelled. Complete the table of information regarding these peaks including the splitting pattern in the **high** resolution spectrum.

(3)

Peak	a	b	c	d
Relative peak area	3			
Chemical shift range, δ / ppm	2.9 – 4.2			
Splitting	singlet			

(ii) State the number of peaks in the carbon-13 (^{13}C) NMR spectrum of **E**.

(1)

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(c) Isomer **F** is a neutral compound that smells of pears.

It can be formed by the reaction between compounds **V** and **W** in the presence of concentrated sulfuric acid.
Compound **V** has an absorption in its infrared spectrum at 1720 cm^{-1} and a broad absorption around 3050 cm^{-1}.
Compound **W** can be formed directly from propanal.
(i) Identify, by name or formula, compounds **V**, **W** and **F**.
Justify your answer by using **all** the available information.

(4)

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(ii) State the conditions required and the reagent needed to form compound **W** from propanal.

(1)

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(iii) State the role of sulfuric acid in the formation of **F**.

(1)

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(d) State which one of the isomers **A**, **E** and **F** has the highest boiling temperature.

Justify your answer.

(3)

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(Total for question = 17 marks)

Q2.

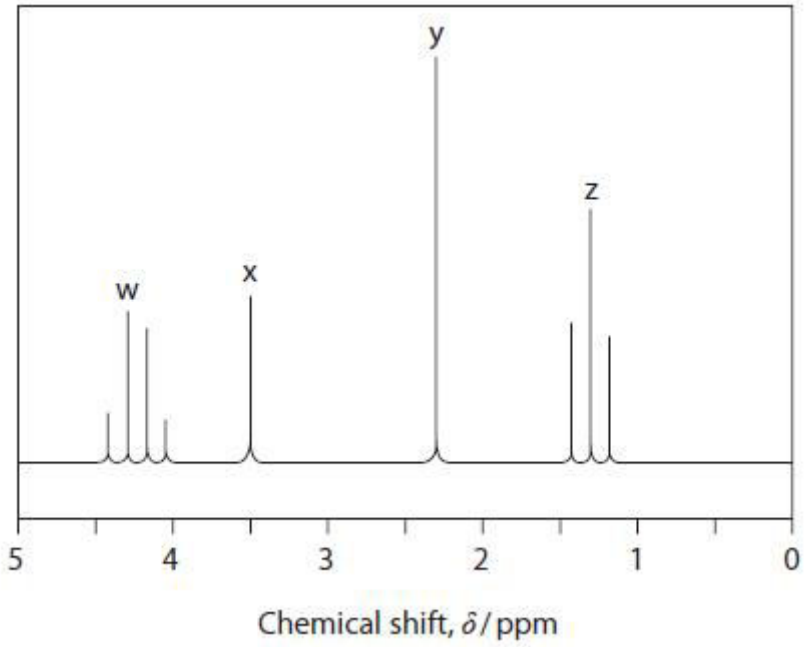
The compound ethyl 3-oxobutanoate, which is used in food flavouring, has the formula $CH_3COCH_2COOCH_2CH_3$.

The high resolution proton (^1H) NMR spectrum of this compound is

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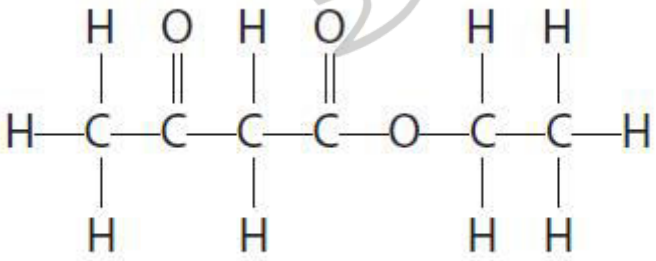
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(a) (i) Identify which groups of hydrogen atoms are responsible for each peak in the spectrum by adding the labels w, x, y and z to the appropriate parts of this displayed formula to match the letters on the spectrum.

(2)



(ii) Explain the splitting patterns of the peaks at 4.2 ppm and 1.3 ppm.

(2)

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(iii) The carbon-13 (^{13}C) NMR spectrum of ethyl 3-oxobutanoate has six peaks.
Draw the structure of an **isomer** of ethyl 3-oxobutanoate that contains a carboxylic acid group and a ketone functional group, but only has four peaks in its carbon-13 NMR spectrum.

(1)

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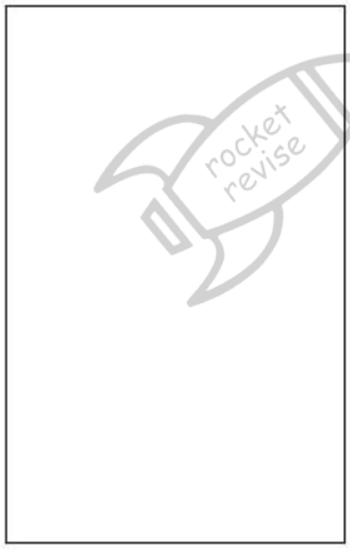
(b) A chemist synthesising ethyl 3-oxobutanoate looked for its presence in the reaction mixture using thin-layer chromatography. The solvent used was a mixture of ethoxyethane, $C_2H_5OC_2H_5$, which is polar, and hexane.

Under these conditions, the R_f value for ethyl 3-oxobutanoate was 0.45.

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(i) Complete the diagram below, showing a chromatogram for ethyl 3-oxobutanoate, including appropriate labels.

(3)



(ii) Suggest why the R_f value for ethyl 3-oxobutanoate is significantly lower than 0.45 when just hexane is used as the solvent.

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(2)

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(Total for question = 10 marks)

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Q3.

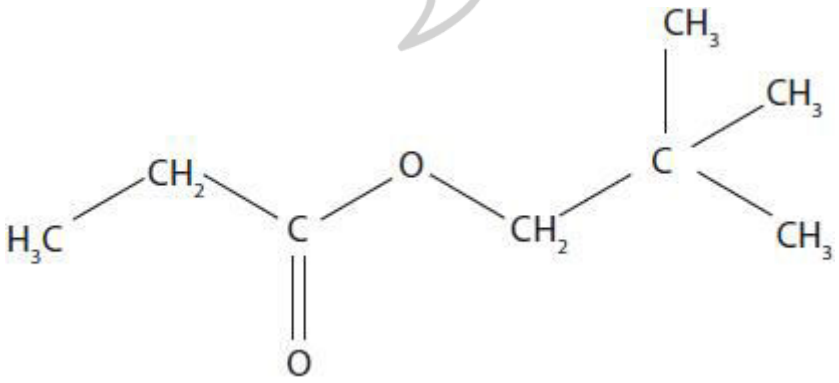
This question is about an ester, **Y**, with the molecular formula $C_8H_{16}O_2$.

(a) **Y** contains 66.7 % carbon, 11.1 % hydrogen and 22.2 % oxygen by mass. Show that these data are consistent with its molecular formula.

(2)

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(b) The structure of compound **Y** is



(i) Give the IUPAC name of **Y**.

(2)

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(ii) Draw the structures of two organic compounds that would react together to form **Y**.

(1)

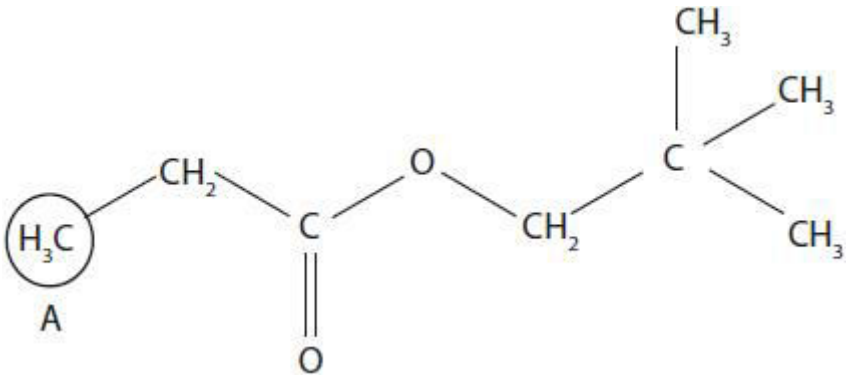
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(c) The high resolution proton NMR spectrum of compound **Y** was obtained.

(i) Label the three remaining hydrogen environments B, C and D on the structure.

(1)



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(ii) Complete the table.

(3)

Hydrogen environment	Splitting pattern of peak	Relative peak area
A	triplet	3
B		
C		
D		

(Total for question = 9 marks)

Q4.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about chromatography.

(a) A spot caused by an amino acid has moved 42 mm from the baseline of a paper chromatogram. The R_f value for the amino acid under these conditions is 0.62.

What is the distance moved by the solvent?

(1)

- ☐ **A** 680 mm
- ☐ **B** 68 mm
- ☐ **C** 42 mm
- ☐ **D** 26 mm

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(b) In gas chromatography, GC, which of these would be the most suitable carrier gas?

(1)

- ☒ **A** argon
- ☒ **B** hydrogen
- ☒ **C** methane
- ☒ **D** oxygen

(Total for question = 2 marks)

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Q5.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A mixture of organic compounds was analysed using thin-layer chromatography.

The R_f value was 0.92 for one of the components in the mixture.

What can be deduced about the attractions between that component and the stationary and mobile phases?

	Attraction between component and stationary phase	Attraction between component and mobile phase
☒ **A**	strong	strong
☒ **B**	strong	weak
☒ **C**	weak	weak
☒ **D**	weak	strong

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(Total for question = 1 mark)

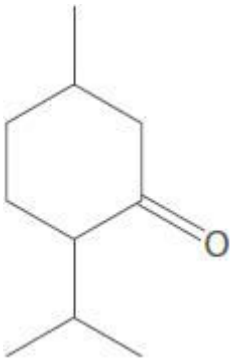
Q6.

This question is about carbonyl compounds.

(a) The skeletal formula of menthone is shown.

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Give the molecular formula of menthone.

(1)

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(b) Ethanal, CH_3CHO , reacts with hydrogen cyanide in the presence of cyanide ions to form 2-hydroxypropanenitrile.

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Draw the mechanism for this reaction.
Include curly arrows, and any relevant lone pairs and dipoles.

(4)

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(c) A carbonyl compound, **F**, has the molecular formula $C_6H_{12}O$.

(i) **F** reacts with iodine in an alkaline solution to give a pale yellow precipitate.
Give the name or formula of the group in **F** identified by this test.

(1)

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(ii) Draw the **skeletal** formulae of the four possible structures of carbonyl compound **F**.

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(2)

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(iii) The carbon-13 (^{13}C) NMR spectrum of **F** has four peaks.
Identify **F** by drawing its **displayed** formula.
Justify your answer by labelling the carbon atoms or groups of carbon atoms responsible for the four peaks in the spectrum.

(2)

*(d) Explain, in terms of all the intermolecular forces involved, why butanal has a higher boiling temperature than pentane but a lower boiling temperature than propanoic acid.

Substance	Boiling temperature / °C
butanal	76
pentane	36
propanoic acid	141

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(6)

(Total for question = 16 marks)

Q7.

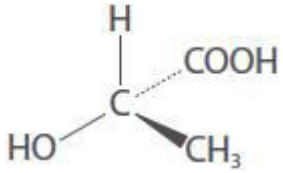
This question is about carboxylic acids and their derivatives.

(a) Lactic acid, $CH_3CH(OH)COOH$, is produced in muscles as a result of anaerobic respiration.

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(i) The structure of lactic acid is



Give a reason why lactic acid shows optical isomerism.

(1)

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(ii) A laboratory sample of lactic acid does **not** rotate the plane of plane-polarised monochromatic light.

Give a reason for this observation.

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(1)

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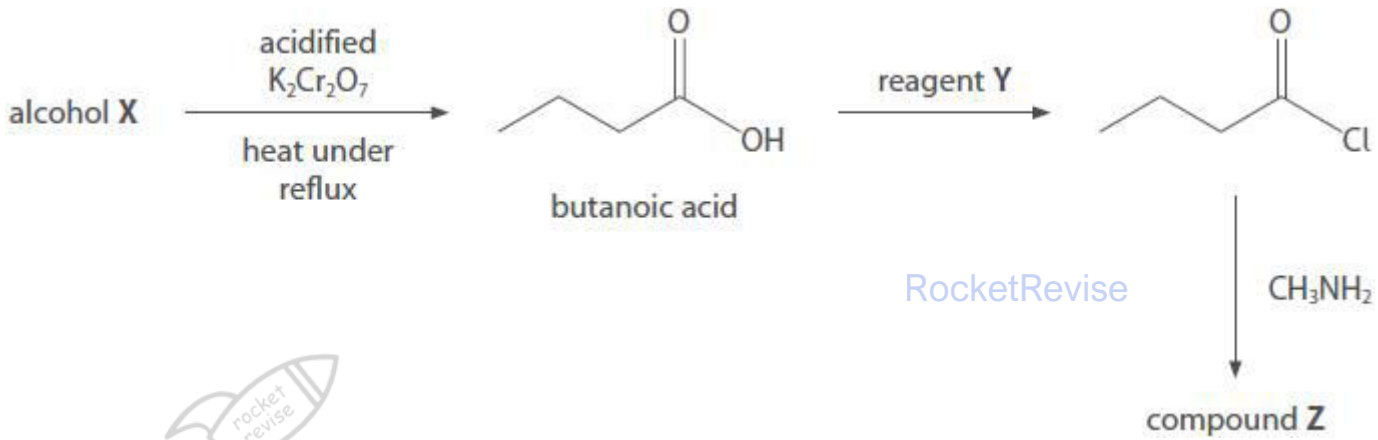
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(iii) Give the structure of the organic product formed when lactic acid reacts with concentrated phosphoric(V) acid, H_3PO_4 .

(1)

(b) A reaction scheme involving butanoic acid is shown.



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Identify **X**, **Y** and **Z** by name or formula.

(3)

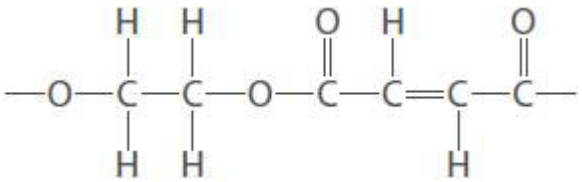
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Alcohol **X**
Reagent **Y**
Compound **Z**

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(c) The repeat unit of a polyester is shown.



Give the structures of the two monomers that could form this polyester.

(2)

Monomer 1	Monomer 2

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(d) An organic compound **E** contains carbon, hydrogen and oxygen only.

(i) The accurate relative atomic masses, A_r , of the three elements in **E** are shown in the table.

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Element	A_r
hydrogen	1.0078
carbon	12.0000
oxygen	15.9949

E contains five carbon atoms and gives a molecular ion peak at $m / z = 102.0678$ in its mass spectrum.
Deduce the molecular formula of **E**.

(1)

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(ii) Aqueous sodium hydrogencarbonate is added to a sample of **E**.
No effervescence occurs.
State what can be deduced by this observation.

(1)

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(iii) The infrared spectrum of **E** has an absorption in the range 1750 – 1735 cm^{-1}.
Name the functional group in **E**.

(1)

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(iv) Data from the high resolution proton NMR spectrum of **E** is shown.

Peak	Chemical shift, δ / ppm for TMS	Splitting pattern	Relative peak area
A	4.02	triplet	2
B	2.05	singlet	3
C	1.65	sextet	2
D	0.95	triplet	3

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Deduce the structure of **E**.
Justify your answer by labelling the protons responsible for each peak.

(3)

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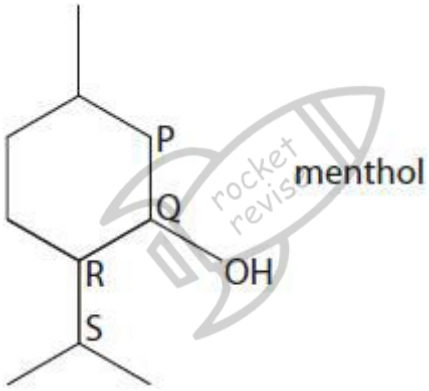
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(Total for question = 14 marks)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The compound menthol has the structure shown.
Some of the carbon atoms are labelled P, Q, R and S.



(a) What is the number of chiral centres in a molecule of menthol?

(1)

- ☐ **A** 1
- ☐ **B** 2
- ☐ **C** 3
- ☐ **D** 4

(b) Which of the carbon atoms is responsible for a peak at 72 ppm in the ^{13}C NMR spectrum of menthol?

(1)

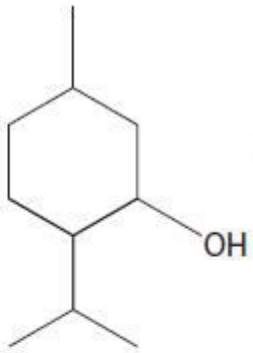
- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

(c) Four groups of students warmed samples of menthol with sodium dichromate(VI) in acid. They purified the reaction mixture and carried out a series of qualitative tests on the organic product.

The findings of each group in the class are shown in the table.

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menthol

Group	Qualitative test		
	Add 2,4-dinitrophenylhydrazine	Warm with Fehling's solution	Add PCl_5
One	✓	×	✓
Two	✓	×	×
Three	✓	✓	×
Four	×	×	✓

A tick (✓) shows a positive result, a cross (✗) shows a negative result.
Which group recorded the results you would expect?

(1)

- ☒ **A** One
- ☒ **B** Two
- ☒ **C** Three
- ☒ **D** Four

(Total for question = 3 marks)

Q9.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The high resolution mass spectrum of a compound X has a molecular ion peak at *m* / *z* = 44.0632.
Accurate relative atomic masses are given in the table.

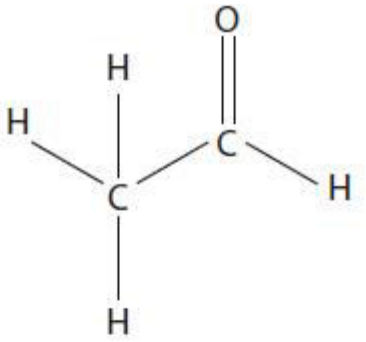
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Element	Relative atomic mass
Hydrogen	1.0079
Carbon	12.0000
Oxygen	15.9949

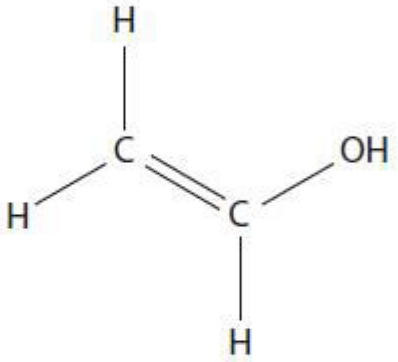
Which of these compounds, with a relative molecular mass of 44, gives rise to this peak?

☐ **A**

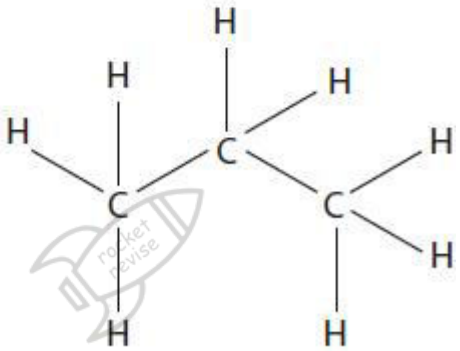


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☐ **B**

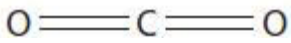


☐ **C**



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☐ **D**



(Total for question = 1 mark)

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Q10.

The standard enthalpy change of solution for ammonium nitrate, NH_4NO_3 , is +25.7 kJ mol^{-1}.

(a) Calculate the value for the standard entropy change in the surroundings, $\Delta S^{\ominus}_{surroundings}$, when ammonium nitrate dissolves in water at 298 K. Include a sign and units with your answer.

(2)

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(b) Explain what can be deduced from your answer in (a) about the sign and the value of the standard entropy change in the system, $\Delta S^{\ominus}_{system}$, when NH_4NO_3 dissolves.

(3)

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(Total for question = 5 marks)

Q11.

Compound **X** is used by mammals as an alternative energy source to sugars.
X is a compound of carbon, hydrogen and oxygen only.

(a) Complete combustion of a 2.50 g sample of **X** in dry oxygen produced 4.31 g of carbon dioxide and 1.32 g of water as the only products.

(i) Give a reason why the oxygen used must be dry.

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(1)

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(ii) Show that the empirical formula of **X** is $C_4H_6O_3$. You **must** show your working.

(5)

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(b) Compound **X** gave an orange precipitate with Brady's reagent (2,4-dinitrophenylhydrazine) but no reaction with Tollens' reagent.

When **X** was added to a solution of sodium hydrogencarbonate, effervescence occurred and the gas evolved turned limewater cloudy.

The carbon-13 NMR spectrum of **X** had only four peaks.

(i) Deduce the **two** possible structures of **X**, showing how this information supports your answer.

(6)

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(ii) Give a **chemical** test which would allow you to distinguish between the two compounds you have given in (b)(i). Include the reagents required and the result for each of the compounds.

(3)

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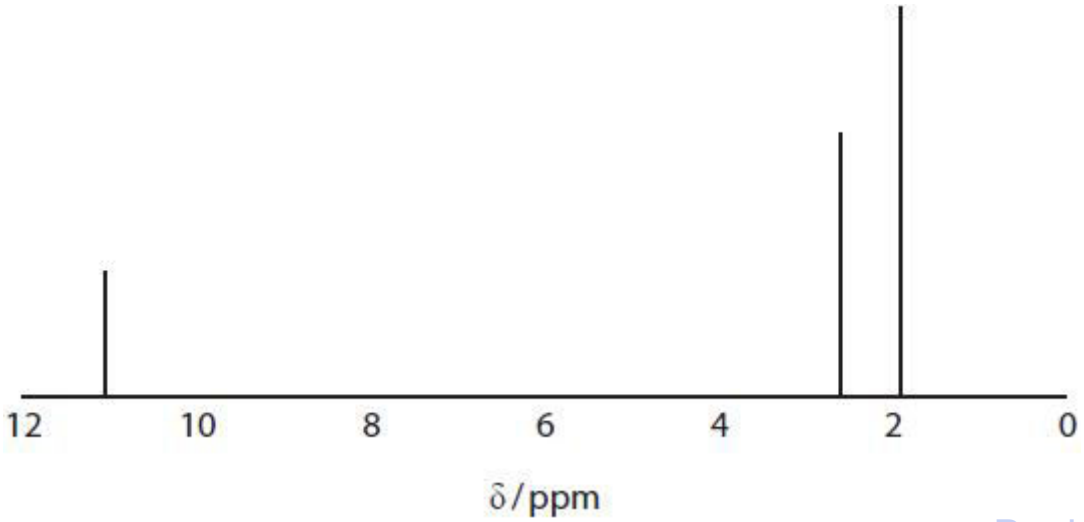
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(c) A simplified **high** resolution proton (^{1}H) NMR spectrum of compound **X** is shown.

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Explain how the number of peaks in the ^{1}H NMR spectrum, together with their relative heights, their chemical shifts and their splitting patterns, may be used to confirm the structure of **X**. Use the chemical shifts given in your Data Booklet.

(5)

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(Total for question = 20 marks)

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Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The high resolution mass spectrum of an organic compound **X** has a molecular ion peak with m / z = 58.0417 and another peak with m / z = 29.0390.

Which compound is **X**?

[A_r values: H = 1.0078 C = 12.0000 O = 15.9949]

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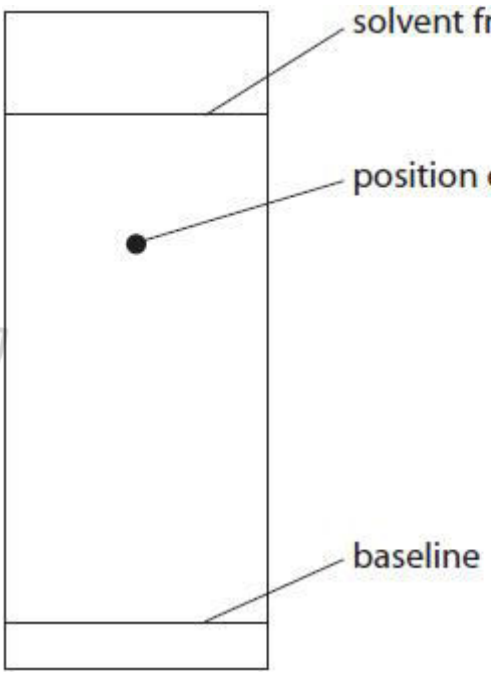
- ☐ **A** CH_3COCH_3
- ☐ **B** CH_3CH_2CHO
- ☐ **C** $CH_3CH_2CH_2CH_3$
- ☐ **D** $CH_3CH(CH_3)CH_3$

(Total for question = 1 mark)

Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A diagram of a thin layer chromatogram of substance **Y** is shown.



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What is the R_f value of substance **Y**?

- ☐ **A** 0.25

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- **B** 0.60
- **C** 0.75
- **D** 1.33

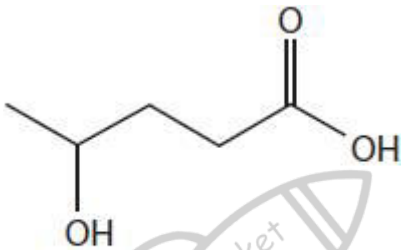
(Total for question = 1 mark)

Q14.

This question is about reactions of carboxylic acids and their derivatives.

(a) Molecule **A** reacts in two ways in the presence of an acid catalyst.

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molecule **A**

Reaction **1** is intramolecular, in which one molecule of **A** forms a cyclic ester **B**.
Reaction **2** is intermolecular, in which a condensation polymer **C** forms.

(i) Draw the structure of the cyclic ester **B**.

(1)

(ii) Draw the structure of the condensation polymer **C**, showing **two** repeat units.

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(2)

(iii) Explain why the entropy change of the system, ΔS_{system}, is greater for Reaction **1** than for Reaction **2**.

(2)

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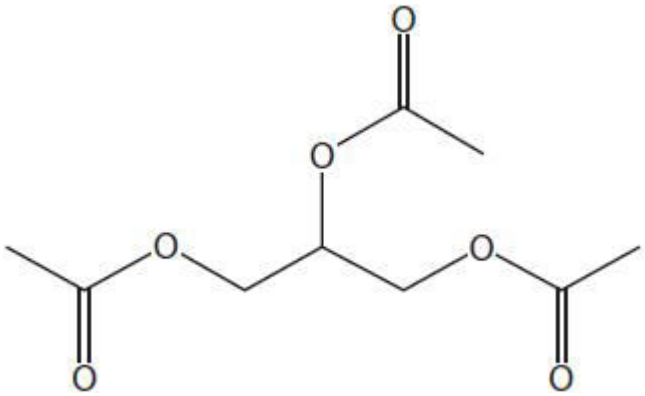
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(b) The molecule shown is a triester, used as a food additive.



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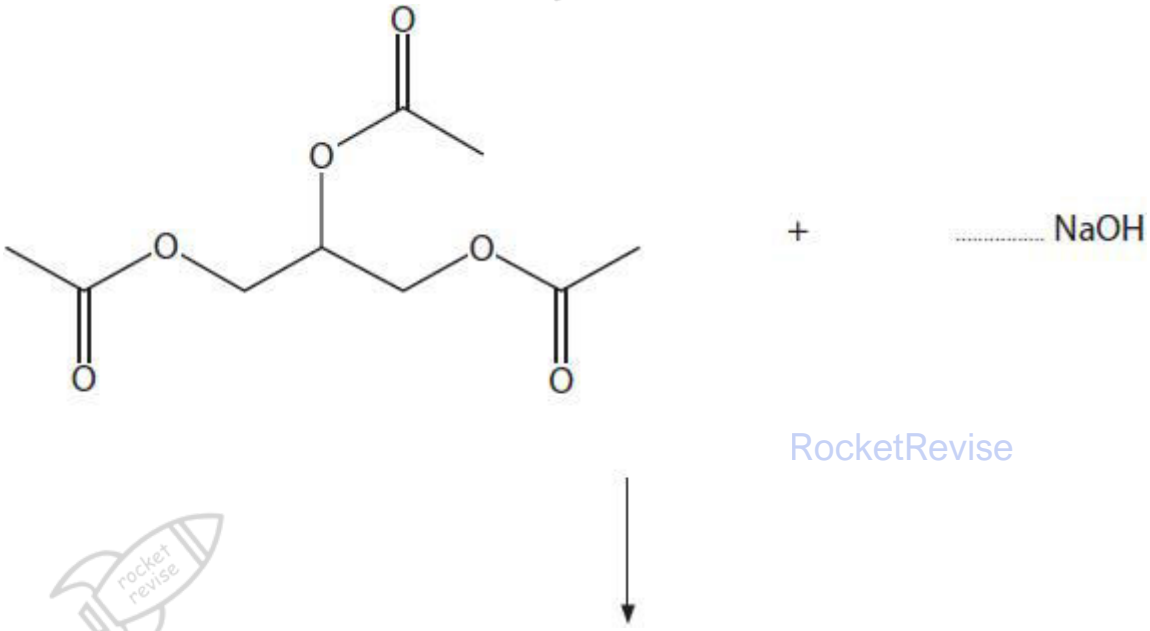
(i) Give the **molecular** formula of this triester.

(1)

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(ii) Complete the equation for the alkaline hydrolysis of this triester with **excess** aqueous sodium hydroxide, showing the structures of the products.

(3)

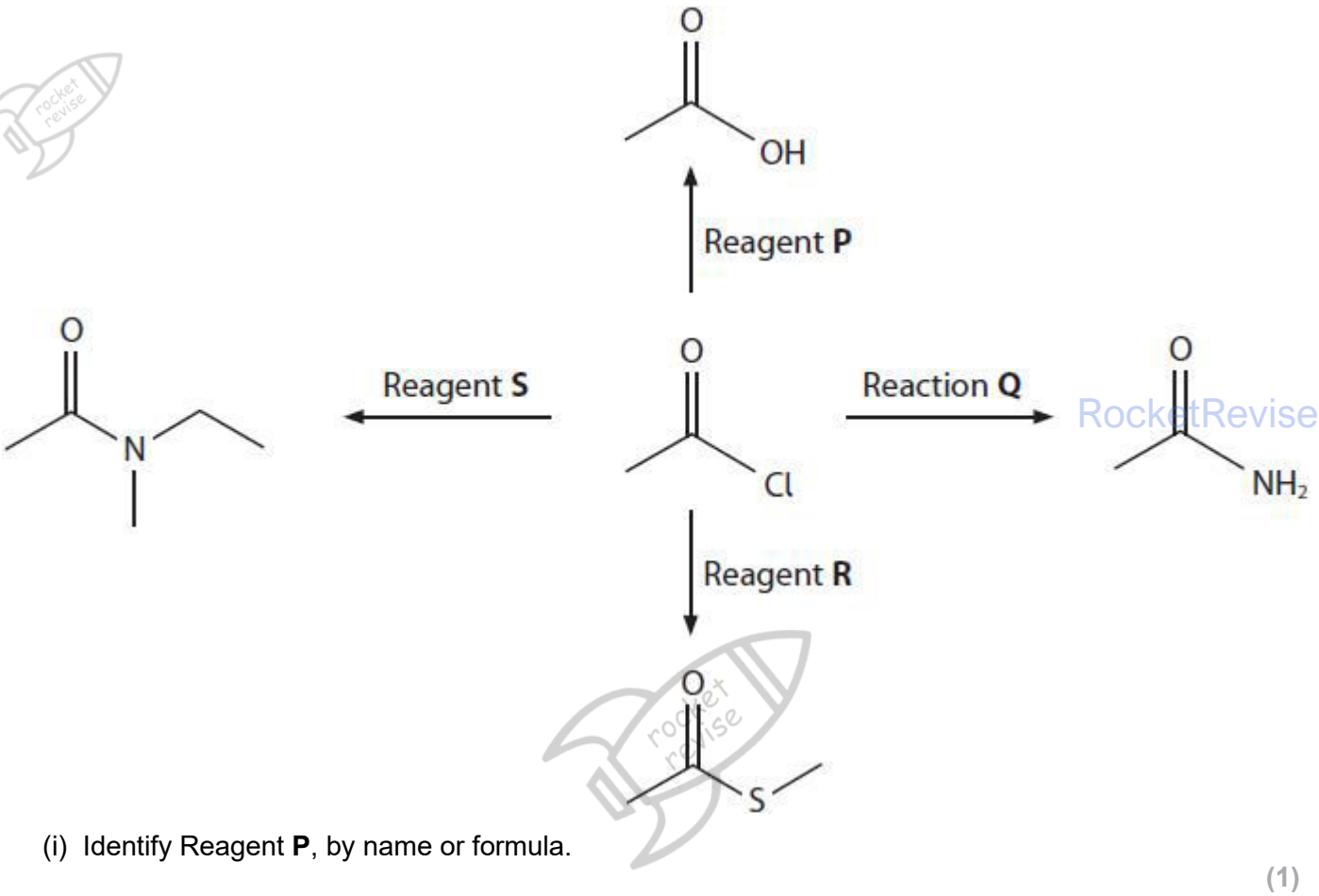


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(c) Some reactions of ethanoyl chloride, CH_3COCl, are shown.



(i) Identify Reagent **P**, by name or formula.

(1)

.....

(ii) Write a balanced equation for Reaction **Q**.
State symbols are **not** required.

(2)

(iii) Suggest the structural formula of Reagent **R**.

(1)

.....

(iv) Draw the **skeletal** formula of Reagent **S**.

(1)

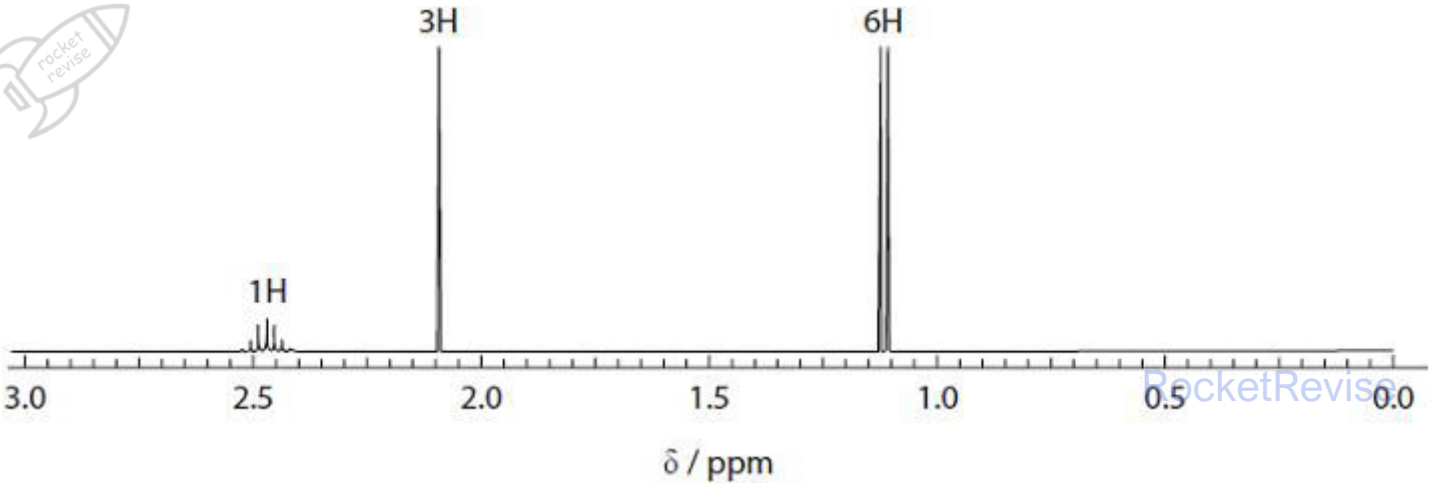
*(d) Sodium ethanoate, CH_3COONa , reacts with 2-methylpropanoyl chloride, $(CH_3)_2CHCOCl$, to form organic compound **Z** and sodium chloride, NaCl.

The molecular ion peak in the mass spectrum of **Z** is at $m / z = 130$.

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The infrared spectrum of **Z** has peaks at 1820 cm^{-1} and 1754 cm^{-1}.
The carbon-13 NMR spectrum of **Z** has **five** peaks.
A simplified high resolution proton NMR spectrum of **Z** is shown. The number of protons in each environment is indicated on the spectrum.



Deduce the structure of **Z**, showing how your answer is consistent with **all** the information provided.

(6)

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(Total for question = 20 marks)

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Q15.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about proton NMR spectroscopy.

(a) How many peaks are there in the **low** resolution proton NMR spectrum of 2-methylpropan-2-ol?

(1)

- ☐ **A** one
- ☐ **B** two
- ☐ **C** three
- ☐ **D** four

(b) Which compound has **three** peaks with relative peak areas of 3 : 2 : 1 in its **low** resolution proton NMR spectrum?

(1)

- ☐ **A** propanal
- ☐ **B** propane
- ☐ **C** propan-1-ol
- ☐ **D** propan-2-ol

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(c) Which compound does **not** have a singlet peak in its **high** resolution proton NMR spectrum?

(1)

- ☐ **A** butanoic acid
- ☐ **B** butanone
- ☐ **C** butanal
- ☐ **D** butan-1-ol

Use this space for any rough working. Anything you write in this space will gain no credit.

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(Total for question = 3 marks)

Q16.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In HPLC, the mobile and stationary phases are

	Mobile phase	Stationary phase
☒ A	gas	liquid
☒ B	gas	solid
☒ C	liquid	gas
☒ D	liquid	solid

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(Total for question = 1 mark)

Q17.

This question is about compounds with the molecular formula $C_6H_{12}O_2$.

(a) Hexanoic acid, $C_5H_{11}COOH$, is a weak acid.

(i) Write the equation for the acid dissociation constant, K_a, of hexanoic acid.

(1)

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(ii) Calculate the pH of a 0.100 mol dm^{-3} solution of hexanoic acid.
[pK_a of hexanoic acid = 4.88]

(4)

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(iii) The solubility of hexanoic acid in water is 1.08 g per 100 g of water.
The isomer of hexanoic acid, butyl ethanoate, $CH_3CO_2C_4H_9$, has a solubility of 0.68 g per 100 g of water.
Explain the differences in these data in terms of the hydrogen bonding between hexanoic acid and water, and between butyl ethanoate and water.

(3)

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(b)(i) Compound **A** is thought to be another isomer of hexanoic acid.

10 g of compound **A** is found to contain 6.21 g of carbon and 1.03 g of hydrogen, with the remainder being oxygen.
Use the data to calculate the empirical formula of compound **A**.
You must show **all** your working.

(3)

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(ii) State how you might use your answer to (b)(i) and a mass spectrum of compound **A** to prove that compound **A** is an isomer of hexanoic acid.

(1)

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*(iii) A series of tests was performed on compound **A**.

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	Test	Observation
1	addition of phosphorus(V) chloride	misty fumes
2	addition of 2,4-dinitrophenylhydrazine	orange precipitate
3	addition of Benedict's or Fehling's reagent	solution remains blue with no precipitate
4	addition of acidified potassium dichromate(VI)	solution remains orange
5	test using polarimetry	plane of plane polarised light is rotated

Deduce the structure for compound **A**.
Justify your answer by using all the test results.

(6)

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(c) Compound **B**, another isomer with the molecular formula $C_6H_{12}O_2$, contains a ring of six carbon atoms.

The carbon-13 NMR spectrum has only two peaks, one of which is at 69 ppm.
Draw the structure of compound **B**.

(2)

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(Total for question = 20 marks)

Q18.

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A high performance liquid chromatography (HPLC) system used a column with a non-polar stationary phase and a solvent of methanol and water.

A mixture of octane, octanal and octan-1-ol was placed into the HPLC column.

What is the order of retention times?

	Shortest retention time	Intermediate retention time	Longest retention time
☒ **A**	octane	octanal	octan-1-ol
☒ **B**	octanal	octan-1-ol	octane
☒ **C**	octane	octan-1-ol	octanal
☒ **D**	octan-1-ol	octanal	octane

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Use this space for any rough working. Anything you write in this space will gain no credit.

(Total for question = 1 mark)

Q19.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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In gas chromatography, a gas containing a mixture is passed over a liquid stationary phase. The main reason a mixture is separated into its components is because they have different.

- ☒ **A** boiling temperatures
- ☒ **B** forces of attraction to the liquid
- ☒ **C** relative molecular masses
- ☒ **d** volatilities

(Total for question = 1 mark)

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Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which change **decreases** the retention time in gas chromatography?

- ☒ **A** decreasing the column temperature
- ☒ **B** increasing the carrier gas flow rate
- ☒ **C** decreasing the amount of sample
- ☒ **D** increasing the column length

(Total for question = 1 mark)

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Q21.

This question is about carbonyl compounds.

(a) Three carbonyl compounds, **A**, **B** and **C**, are **straight-chain** structural isomers, with the formula $C_5H_{10}O$.

Only isomer **A** reacts with Tollens' reagent to give a silver mirror.
Only isomer **B** reacts with iodine in the presence of alkali to produce pale yellow crystals.
Draw the **displayed** structures of these three isomers.

(3)

A	
B	
C	

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(b) Another carbonyl compound, propanal, reacts with HCN in the presence of KCN to form a racemic mixture of two optical isomers of $CH_3CH_2CH(OH)CN$.

(i) Give the IUPAC name for $CH_3CH_2CH(OH)CN$.

(1)

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(ii) Describe how you could distinguish between pure samples of the two optical isomers.

(1)

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(iii) Explain, with reference to the reaction mechanism, why this reaction produces a racemic mixture.

(2)

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(c) (i) Propanone, an isomer of propanal, also reacts with HCN in the presence of KCN.

Draw the **skeletal** formula of the product of this reaction.

(1)

(ii) State why the product formed in (c)(i) does **not** show optical isomerism.

(1)

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(d) ^{13}C NMR spectroscopy provides information about the structures of propanal and propanone.

(i) Identify the chemical shift range and carbon environment of **one** peak you would expect to see in **both** spectra.

(1)

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Chemical shift range	Carbon environment

(ii) State the number of peaks you would expect to see in each ^{13}C NMR spectrum.

(1)

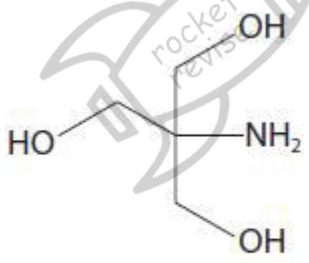
Propanal .....
Propanone .....

(Total for question = 11 marks)

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Q22.

The alkaline compound tris(hydroxymethyl)aminomethane, known as Tris, is used to make a buffer for biological research.



Tris

(a) Sketch the **low** resolution proton NMR spectrum of Tris ($C_4H_{11}NO_3$).

Use your Data Booklet.

(3)

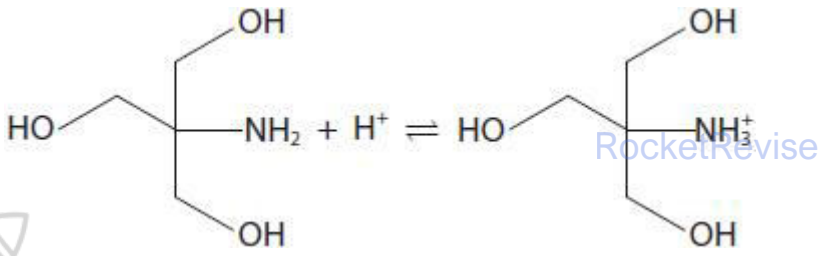
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(b) Tris is a Brønsted‐Lowry base and its conjugate acid is formed as shown.



(i) Explain how a mixture of Tris and its conjugate acid acts as a buffer solution when a small amount of acid is added.

(3)

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(ii) Write the expression for the K_a of the conjugate acid of Tris $(C_4H_{12}NO_3^+)$.

(1)

(iii) When hydrochloric acid is added to Tris, the acid salt is formed.

The acid salt is a solid, which has the formula $C_4H_{12}NO_3^+Cl^-$, and contains the conjugate acid of Tris.

When 100 g of the acid salt is mixed with 500 cm^3 of 0.200 mol dm^{-3} Tris, an alkaline buffer is formed.

Calculate the pH of this buffer, assuming that there is no change in volume when the solid is added.

K_a for the conjugate acid of Tris is 8.413×10^{-9} mol dm^{-3}.

(5)

(c) A solution of chloroethanoic acid is prepared for titration with Tris.

0.0150 g of chloroethanoic acid (M_r = 94.5) is dissolved in 1500 cm^3 of distilled water. The resulting solution has a pH of 3.42.

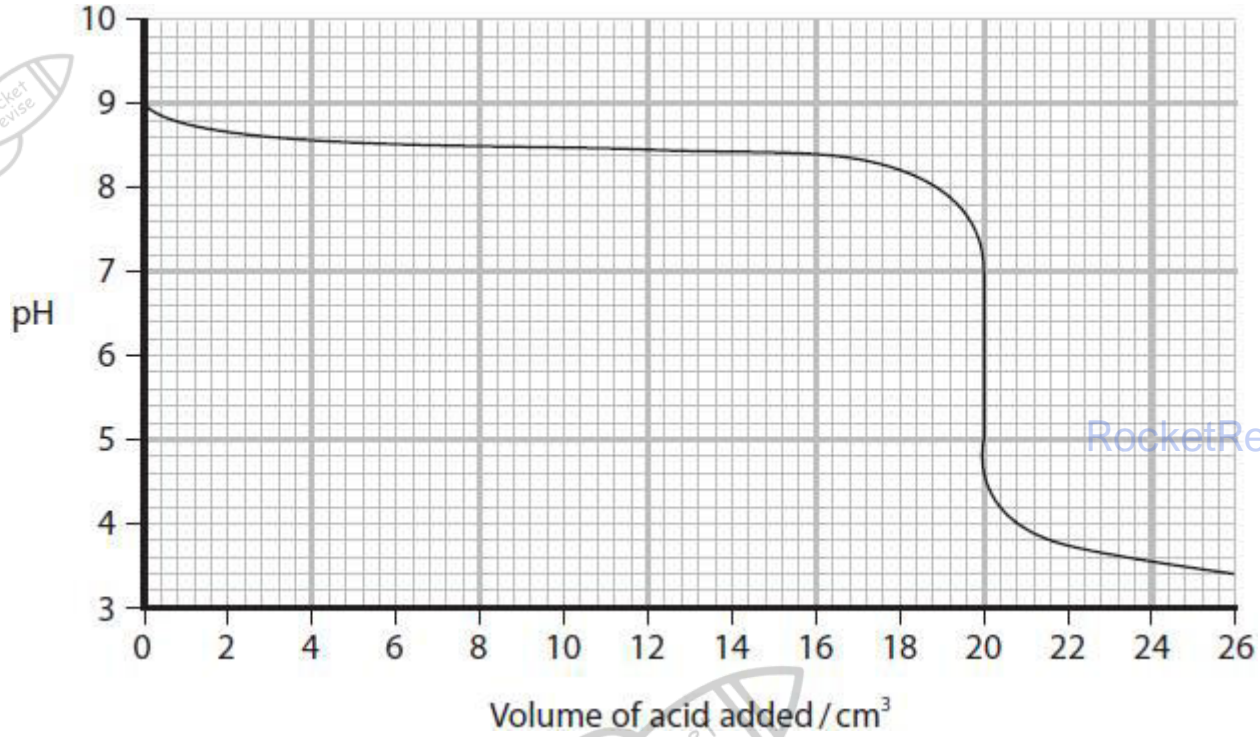
Calculate the K_a of chloroethanoic acid.

(4)

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(d) A titration curve of Tris with chloroethanoic acid is shown.



(i) Explain how this graph shows Tris and its conjugate acid act as a buffer.

(2)

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(ii) Use the graph to estimate the pH of the salt formed when Tris is neutralised with chloroethanoic acid.

(1)

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(iii) Suggest a reason why buffers are so important in biological systems.

(1)

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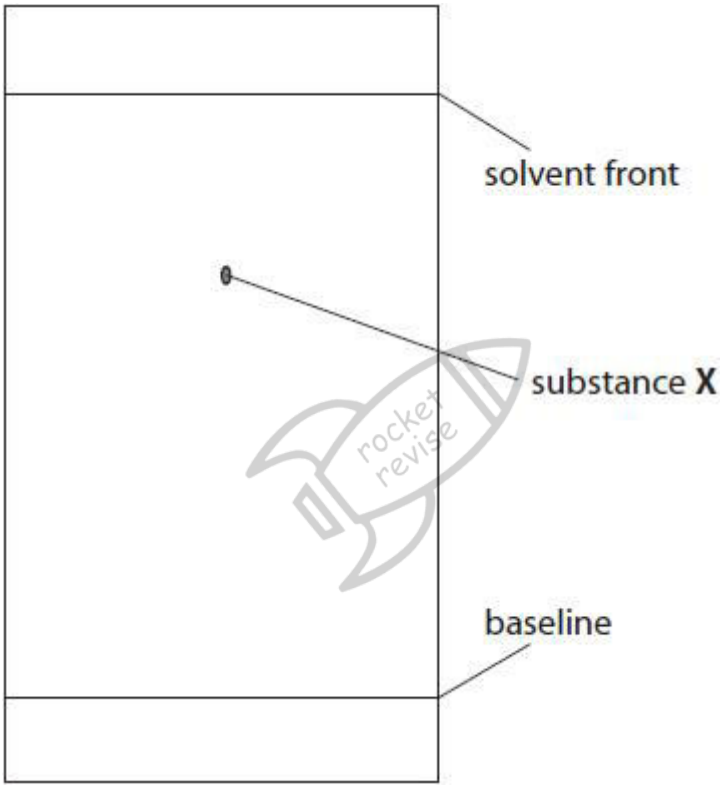
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(Total for question = 20 marks)

Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The chromatogram of a substance **X** is shown.



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What is the R_f value for substance **X**?

- ☒ **A** 0.28
- ☒ **B** 0.62
- ☒ **C** 0.70
- ☒ **D** 0.86

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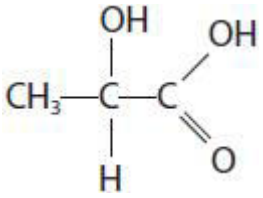
(Total for question = 1 mark)

Q24.

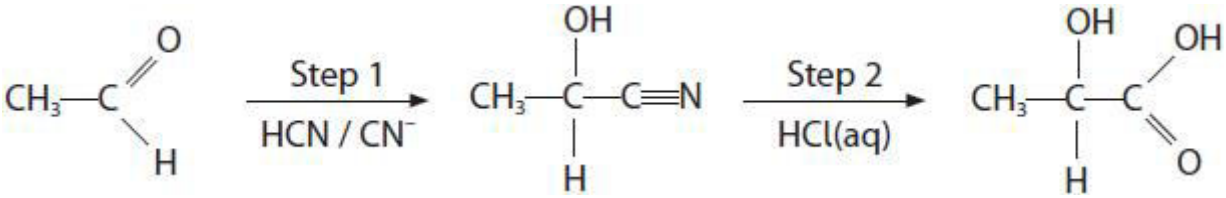
Lactic acid is formed in muscles and in sour milk.

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Lactic acid may be obtained from ethanal in the laboratory by the two-step synthesis shown.



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*(a) Explain why lactic acid formed in muscles is optically active while lactic acid obtained using the laboratory synthesis is not.

You should define any terms you use.

(6)

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(b) Poly(lactic acid) is a polyester derived from lactic acid as the **only** monomer.

Poly(lactic acid), which is often referred to as PLA, may be obtained from renewable resources and is also biodegradable.

(i) Draw the structure of a section of PLA showing **two** repeat units.

(2)

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(ii) Lactic acid is converted into compound **X** which is then polymerised to form PLA.

The empirical formula of **X** is $C_3H_4O_2$ and its molar mass is 144 g mol^{-1}.

Determine the molecular formula of **X**.

(1)

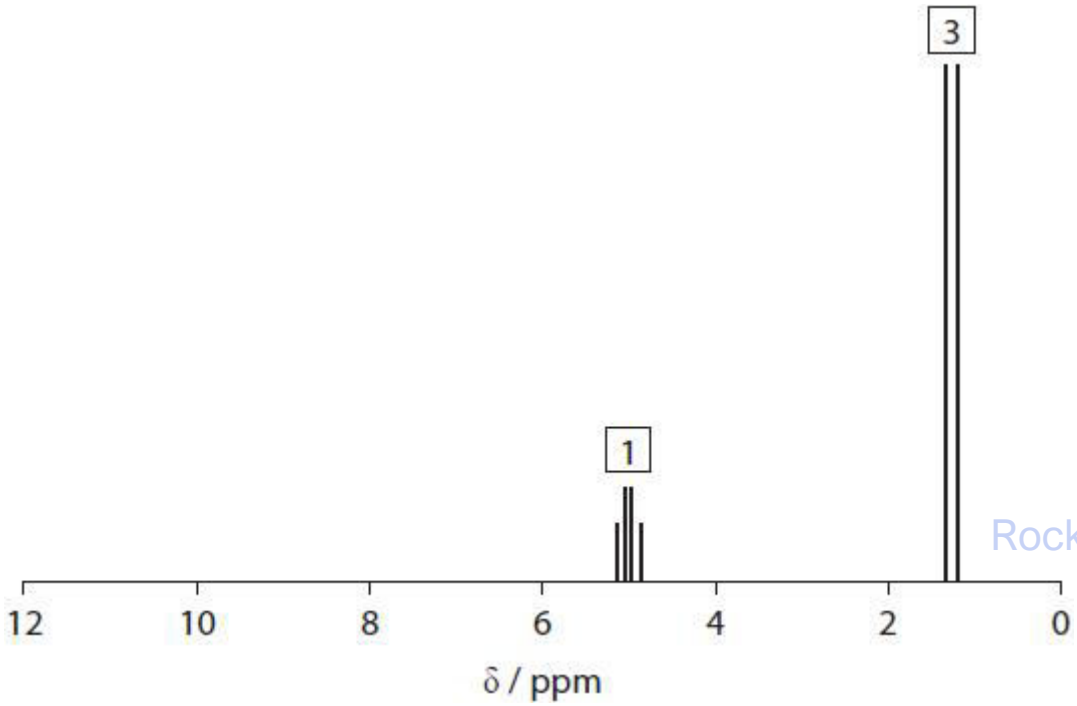
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(iii) A simplified high resolution proton NMR spectrum of **X** is shown. The numbers above the peaks are the relative peak areas.

Note that the chemical shifts for **X** are different from the general values in the Data Booklet.

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State what the number of peaks, their relative peak areas and splitting patterns show about the structure of **X**.

(3)

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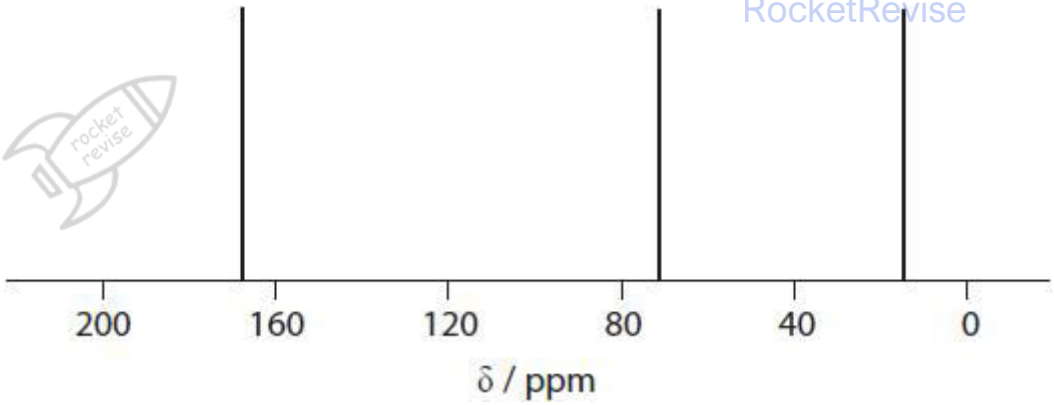
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(iv) A simplified carbon-13 NMR spectrum of **X** is shown.

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State what additional information about the structure of **X** is given by this spectrum.

(2)

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(v) Draw the structure of **X** using the information from (b)(ii) to (b)(iv).

(1)

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(Total for question = 15 marks)

Q25.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

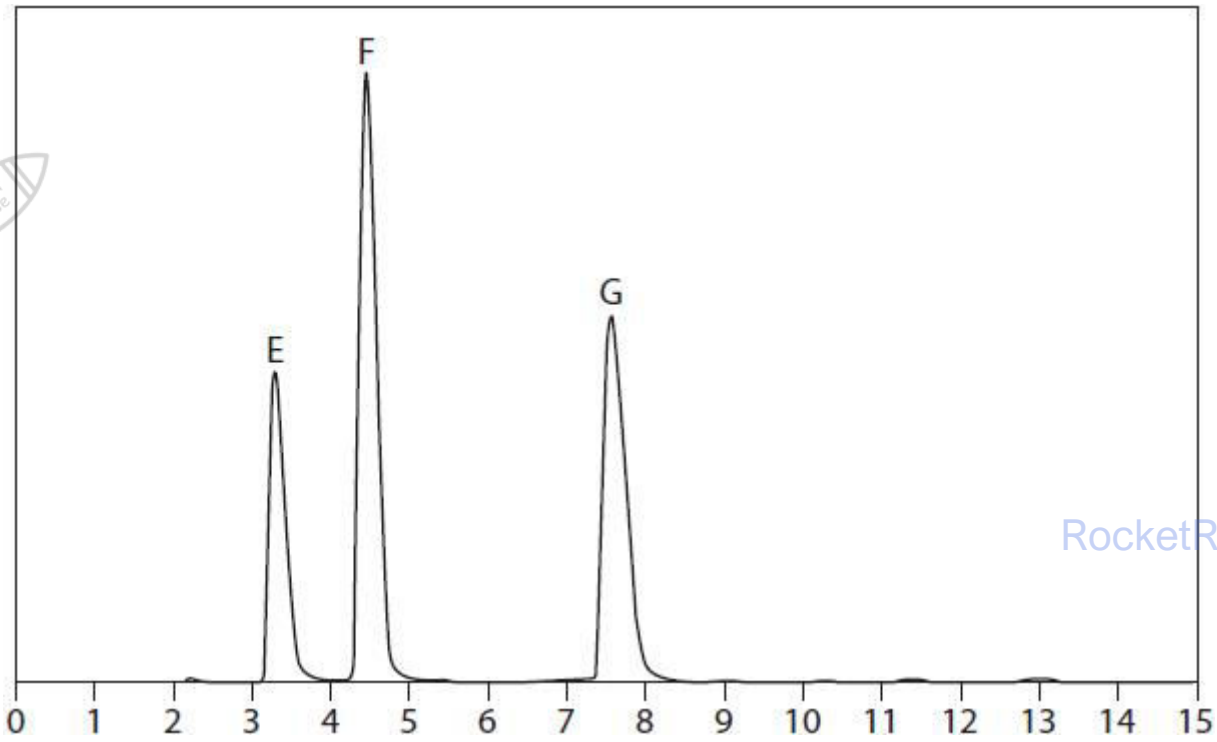
High-performance liquid chromatography (HPLC) is used to separate a mixture into its three components.

The resulting chromatogram is shown.

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(a) Which is correct for the labels on the axes?

(1)

		x-axis	y-axis
☒	**A**	absorption	time
☒	**B**	R_f	absorption
☒	**C**	time	R_f
☒	**D**	time	absorption

(b) Which is correct for the components E, F and G ?

(1)

		Most attracted to stationary phase	Most abundant
☒	**A**	E	E
☒	**B**	G	E
☒	**C**	E	F
☒	**D**	G	F

(Total for question = 2 marks)

Q26.

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Carbonyl compounds are usually reduced in the laboratory using complex metal hydrides such as lithium tetrahydridoaluminate(III), $LiAlH_4$.
The metal hydrides react by supplying hydride ions, H^-, which then react with the carbonyl group to form an intermediate.
The addition of a strong acid in aqueous solution to the intermediate produces the reduction product.

(a) State the essential condition for using $LiAlH_4$.

(1)

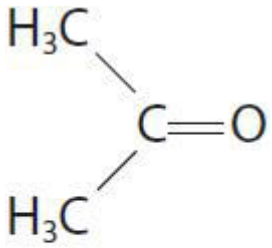
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(b) Complete the mechanism for the reduction of propanone with $LiAlH_4$, showing the structure of the intermediate **and** of the final product.

Include curly arrows, and relevant lone pairs and dipoles.

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(4)





H^-

(c) Explain why $LiAlH_4$ reduces carbonyl compounds but not alkenes, even though both types of compound have π bonds.

(3)

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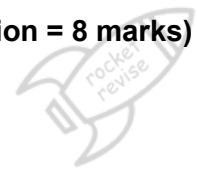


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(Total for question = 8 marks)



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Q27.

This question is about some bromoalkanes.

(a) There are three straight-chain structural isomers with the molecular formula $C_5H_{11}Br$.

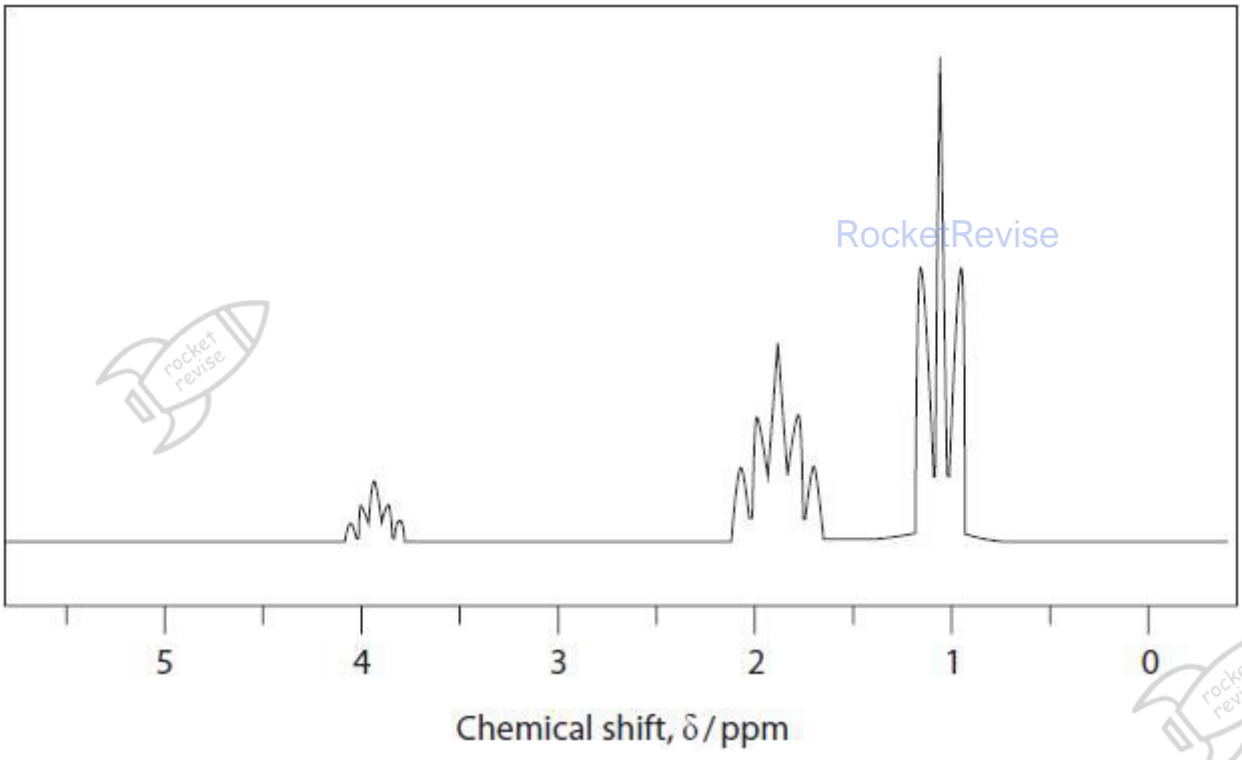
(i) Complete the table for these three isomers.

(3)

Isomer	Skeletal formula	Number of peaks in ^{13}C NMR spectrum
1		.....
2		.....
3		.....

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* (ii) The high resolution proton NMR spectrum of one of these isomers is shown.



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Deduce which isomer is present in the sample, explaining the splitting patterns and chemical shifts seen in the spectrum.

Include the name of the isomer and the relative peak areas.

Use your Data Booklet.

(6)

(b) Draw the S_N2 mechanism for the reaction of 1-bromopropane with hydroxide ions in aqueous solution.

Include curly arrows, and relevant dipoles and lone pairs.

(4)

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(Total for question = 13 marks)

Q28.

In gas chromatography (GC), which one of the following would **increase** the retention time for an alcohol?

(1)

- ☒ **A** increasing the flow rate of the mobile phase
- ☒ **B** increasing the polarity of the stationary phase
- ☒ **C** decreasing the polarity of the stationary phase
- ☒ **D** increasing the polarity of the mobile phase

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(Total for question = 1 mark)

Q29.

One of the compounds responsible for the characteristic smell of bananas is 3-methylbutyl ethanoate. The structural formula of this compound is:

$$CH_3COOCH_2CH_2CH(CH_3)_2$$

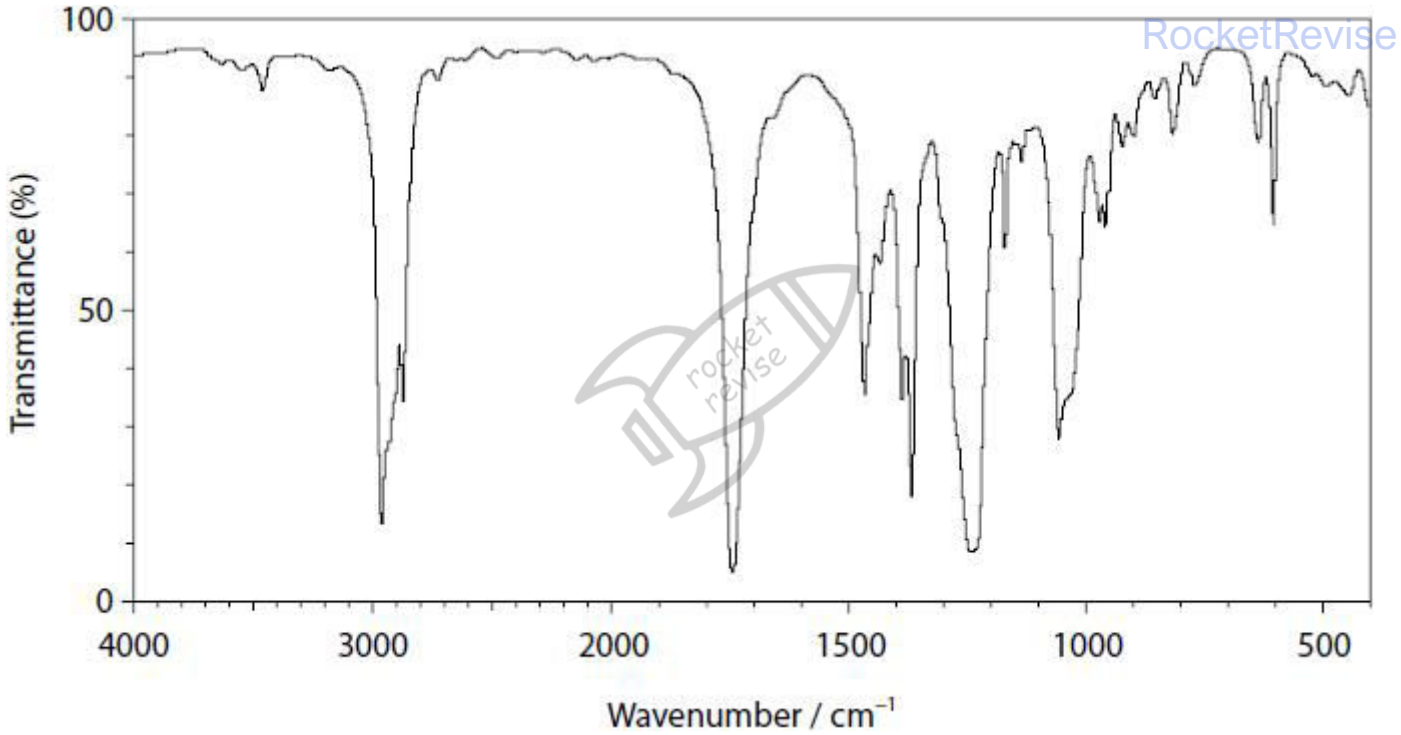
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(a) Draw the **skeletal** formula for 3-methylbutyl ethanoate.

(1)

(b) The infrared spectrum of a sample of 3-methylbutyl ethanoate is shown below.



(i) Use this spectrum to identify one peak resulting from a bond in the ester group of 3-methylbutyl ethanoate. Include the relevant bond and its wavenumber range.

(1)

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(ii) 3-methylbutyl ethanoate was synthesised by reacting a suitable carboxylic acid with an alcohol. Show that its infrared spectrum confirms there is no alcohol remaining in the sample.

(1)

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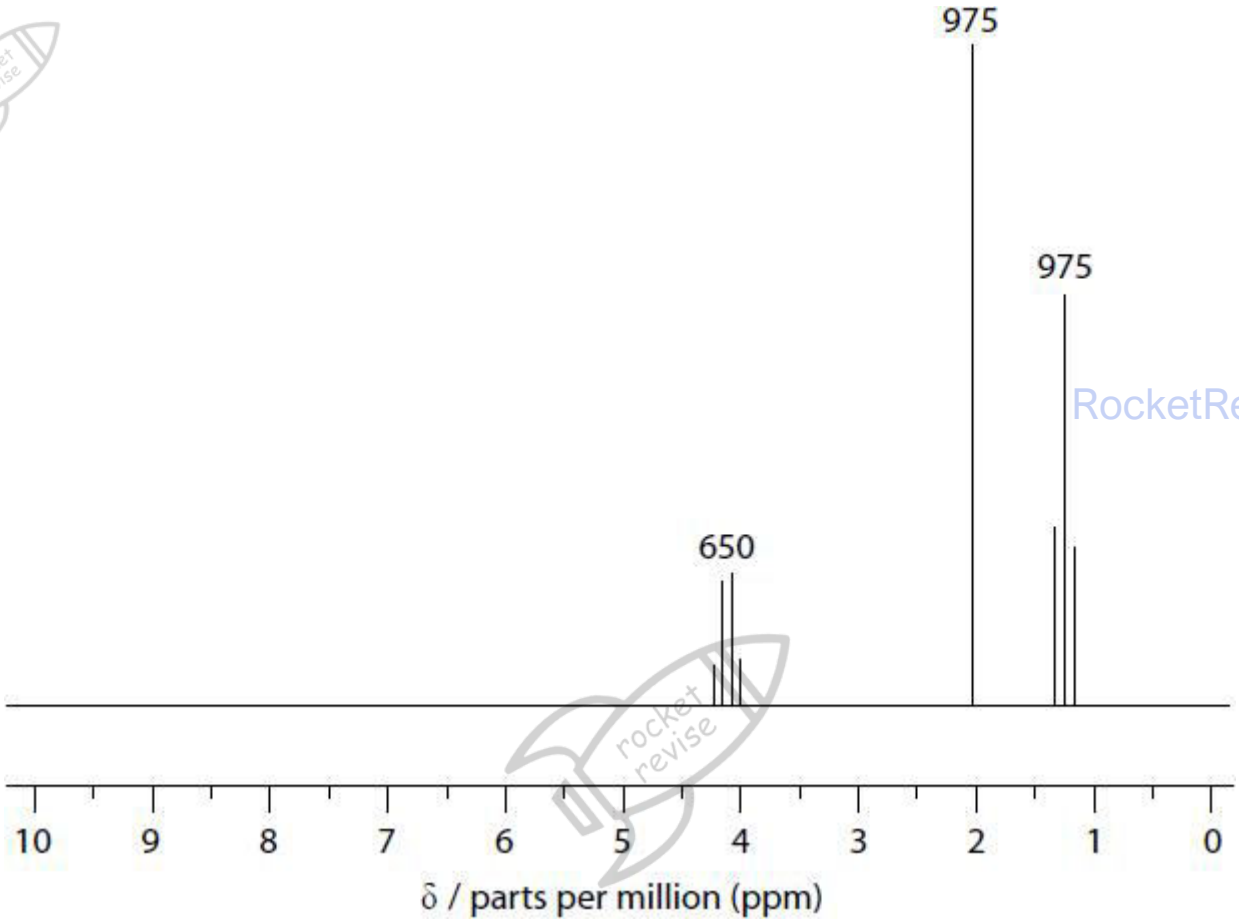
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*(c) The high resolution proton nuclear magnetic resonance (NMR) spectrum of a different ester, ethyl ethanoate, $CH_3COOCH_2CH_3$, is shown.

The numbers over each peak represent their approximate relative areas.



Show that the structure of ethyl ethanoate is consistent with this NMR spectrum, using all the data in the spectrum.

(6)

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(d) Ethyl ethanoate has three isomers which are also esters.

(i) Draw the structures of these three isomers.

(2)

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(ii) Explain to what extent it is possible to distinguish between the three isomers using carbon-13 NMR spectroscopy.

(2)

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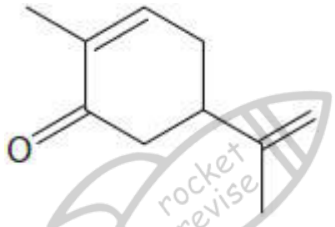
.....

(Total for question = 13 marks)

Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Carvone is an oil used in aromatherapy.



(a) Carvone shows

(1)

- ☐ **A** geometric and optical isomerism
- ☐ **B** geometric isomerism only
- ☐ **C** optical isomerism only
- ☐ **D** neither geometric nor optical isomerism

(b) Which reagent gives a positive result when added to carvone?

(1)

- ☐ **A** ammoniacal silver nitrate (Tollens' reagent)
- ☐ **B** aqueous sodium carbonate
- ☐ **C** iodine in the presence of an alkali
- ☐ **D** 2,4-dinitrophenylhydrazine

(c) How many peaks would be expected to appear in a carbon–13 (^{13}C) NMR spectrum of carvone?

(1)

- ☐ **A** 10
- ☐ **B** 9
- ☐ **C** 8
- ☐ **D** 7

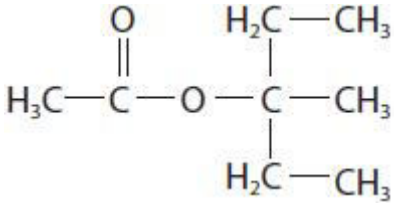
(Total for question = 3 marks)

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Q31.

An ester, **X**, with the molecular formula $C_8H_{16}O_2$, has the structure shown.



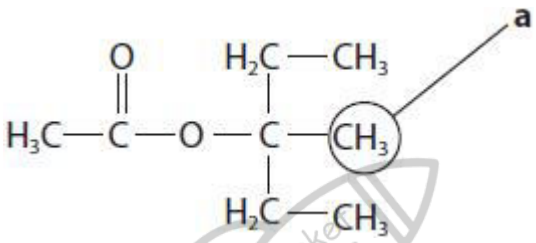
(a) Deduce the name of the alcohol used to synthesise **X** in a reaction with ethanoyl chloride.

(1)

.....

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(b) One of the proton environments in **X** is labelled '**a**'.



(i) Complete the labelling of this structure to show each of the other proton environments in **X**, clearly linking any that are the same.

(2)

(ii) Describe **three** ways in which the high resolution ^1H NMR spectrum can be used to distinguish between the **methyl** groups in **X**, other than **a**.

(3)

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(c) An isomer of **X** is a carboxylic acid with five peaks in its ^{13}C NMR spectrum.

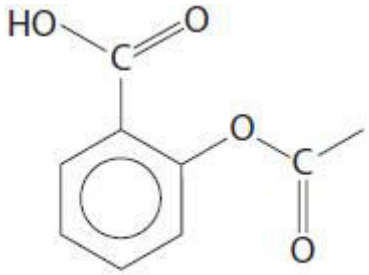
Draw a possible skeletal structure for this carboxylic acid.

(1)

(d) The painkiller aspirin is also an ester, formed by the reaction of 2-hydroxybenzoic acid and ethanoic anhydride.

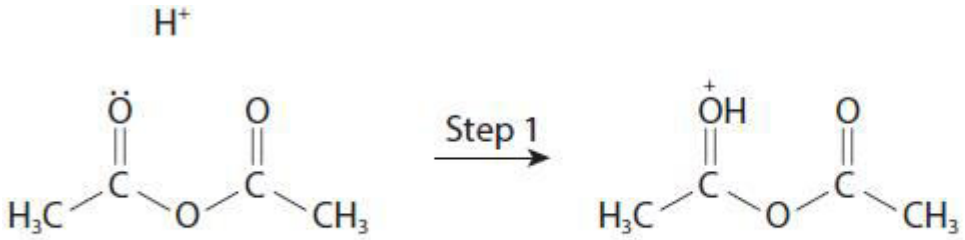
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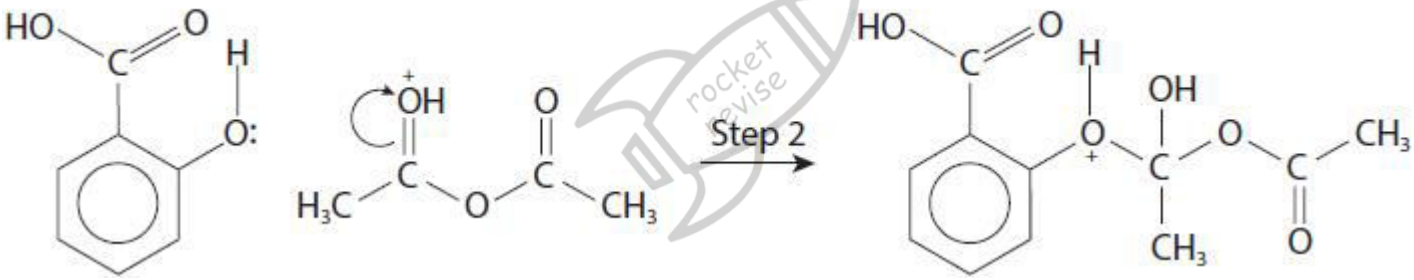


aspirin

Part of the mechanism for this reaction is shown.



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(i) Add a single curly arrow to Step 1 and a single curly arrow to Step 2 to complete this part of the mechanism.

(2)

(ii) State the role of the H^+ ion in the synthesis of aspirin.

(1)

.....

(Total for question = 10 marks)

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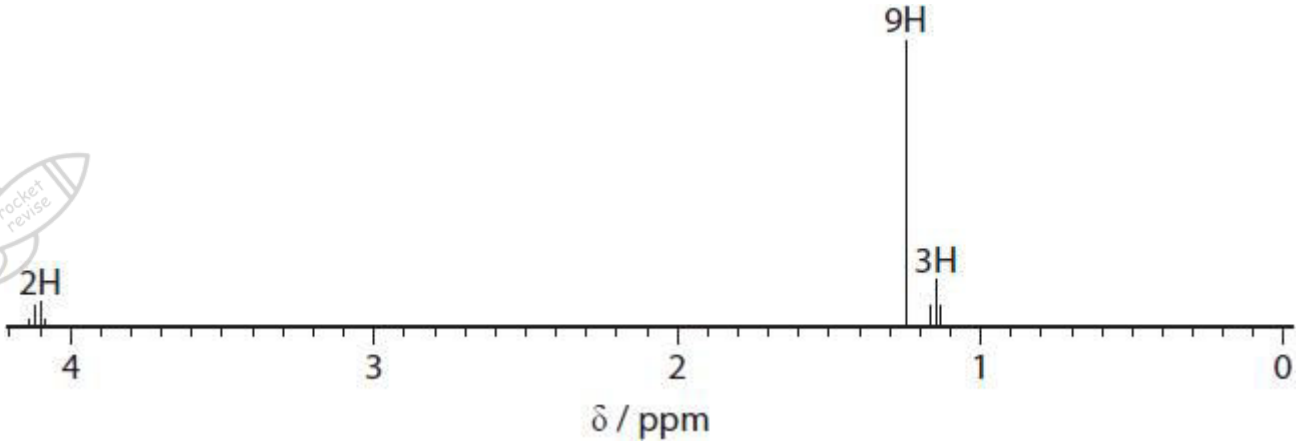
Q32.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The high resolution ^1H NMR spectrum of an ester is shown.

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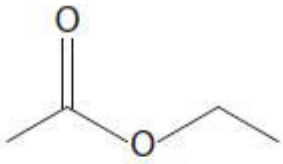
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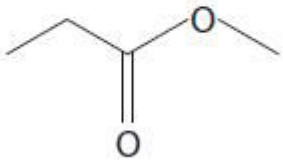
What is the skeletal formula of this ester? Use the Data Booklet as a source of information.

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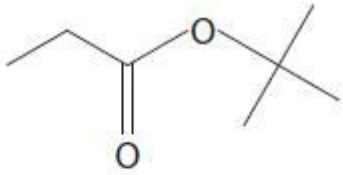
☒ **A**



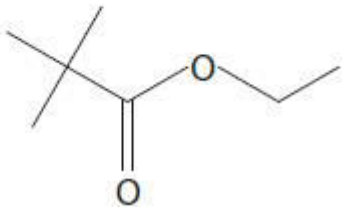
☒ **B**



☒ **C**



☒ **D**



(Total for question = 1 mark)

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Q33.

Methyl 2-hydroxypropanoate is used as an industrial solvent.

It may be prepared in three steps using ethanal as the starting material.

$$CH_3CHO \xrightarrow{\text{Step 1}} CH_3CH(OH)CN \xrightarrow{\text{Step 2}} CH_3CH(OH)COOH \xrightarrow{\text{Step 3}} CH_3CH(OH)COOCH_3$$

(a) Name the reagent(s) in Step **1** and Step **2**.

(2)

Step 1

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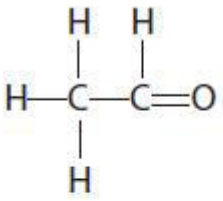
Reagent(s) .....

Step 2

Reagent(s) .....

(b) (i) Complete the mechanism for Step **1**, using curly arrows and relevant lone pairs, charges and dipoles.

(4)



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(ii) Explain why the 2-hydroxypropanoic acid, $CH_3CH(OH)COOH$, produced is **not** optically active.

(3)

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.....

.....

.....

(c) (i) State the type of reaction in Step **3**.

(1)

.....

(ii) A small amount of a polymeric compound is formed during Step **3**. Deduce the structure for the repeat unit of the polymer formed.

(1)

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(d) The high-resolution proton nuclear magnetic resonance (NMR) spectrum of methyl 2-hydroxypropanoate gives four peaks, **J, K, L** and **M**.

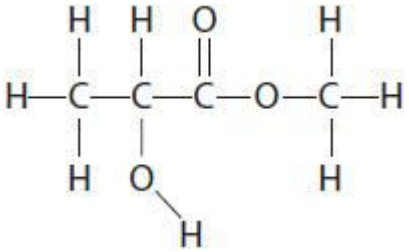
Peaks **L** and **M** are **singlets** with relative intensities of **one** and **three** respectively.

(i) Label the displayed formula to show the protons responsible for these two peaks.

(2)

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(ii) Complete the table to show the expected number of hydrogen atoms and expected splitting pattern for peaks **J** and **K**.

(2)

Peak	δ / ppm	Number of hydrogen atoms	Splitting pattern
J	1.3		
K	4.1		
L	3.6	1	singlet
M	3.7	3	singlet

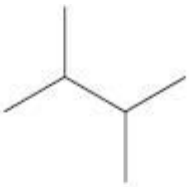
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(Total for question = 15 marks)

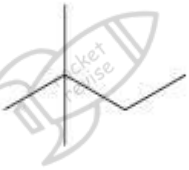
Q34.

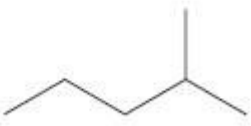
Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which isomer has the most peaks in its ^{13}C NMR spectrum?

☐ **A** 

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☐ **B** 

☐ **C** 

☐ **D** 

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(Total for question = 1 mark)

Q35.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which compound has **most** peaks in its ^{13}C NMR spectrum?

- ☒ **A** CH_3COCH_3
- ☒ **B** $CH_3COCH_2CH_3$
- ☒ **C** $CH_3CH_2COCH_2CH_3$
- ☒ **D** $CH_3COCH_2CH_2COCH_3$

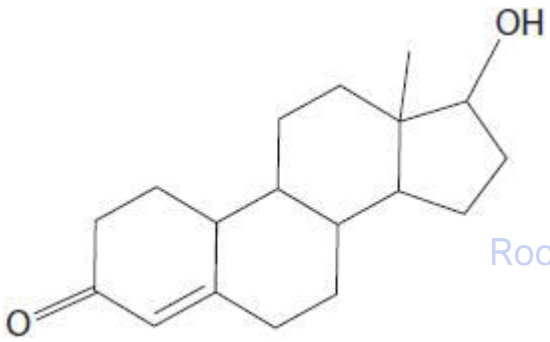
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(Total for question = 1 mark)

Q36.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The steroid nandrolone has the structure shown.



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(a) How many chiral centres are present in nandrolone?

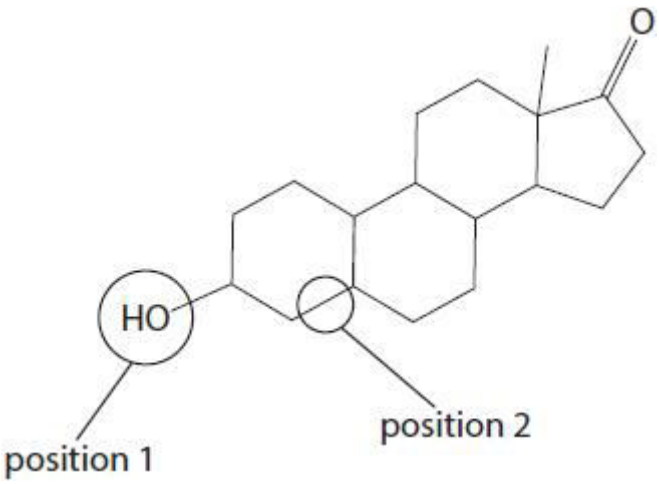
(1)

- ☒ **A** six
- ☒ **B** seven
- ☒ **C** eight
- ☒ **D** nine

(b) Nandrolone is broken down in the body into the compound shown.

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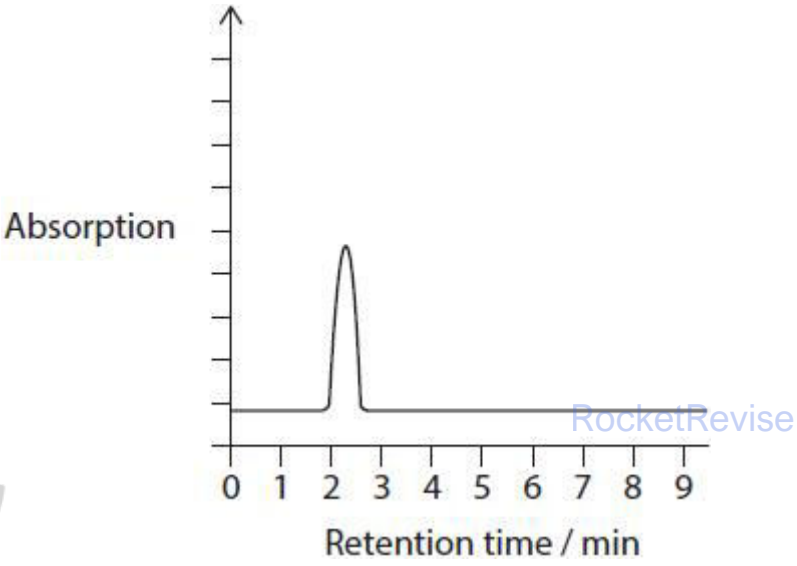


What reactions have taken place to form the new bonds at position 1 and position 2?

(1)

		Position 1	Position 2
☒	A	reduction	substitution
☒	B	reduction	addition
☒	C	oxidation	substitution
☒	D	oxidation	addition

(c) A sample containing 2×10^{-9} g cm^{-3} of nandrolone was analysed using gas chromatography. The chromatogram produced is shown.

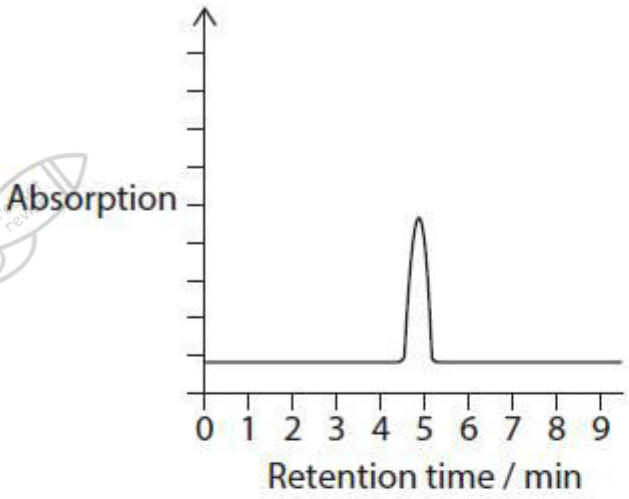


A second sample containing 4×10^{-9} g cm^{-3} of nandrolone was also analysed using gas chromatography, under the same conditions.
Which is the chromatogram produced by this second sample?

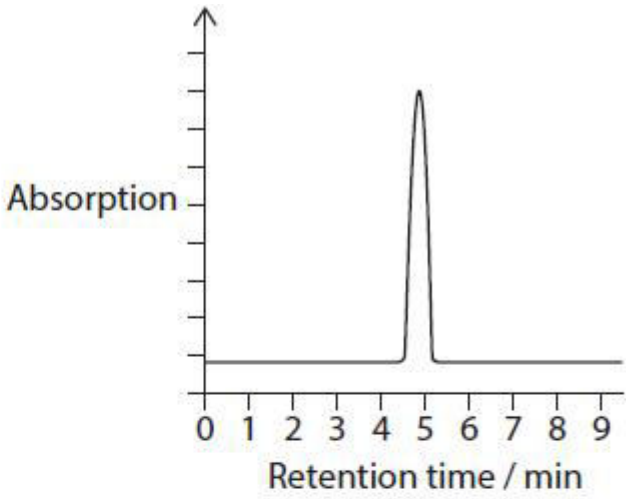
(1)

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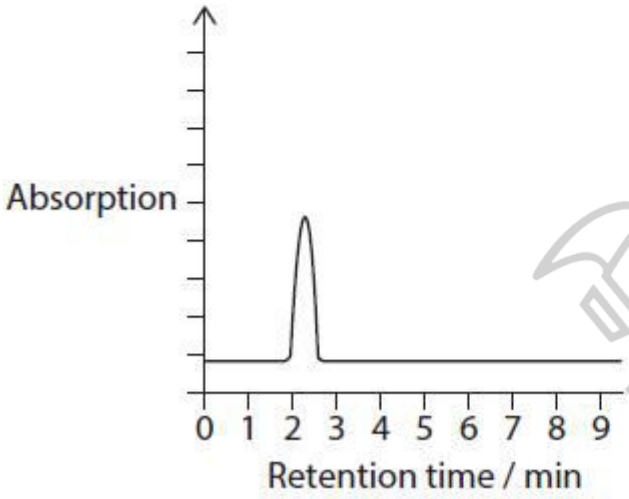


☒ **A**

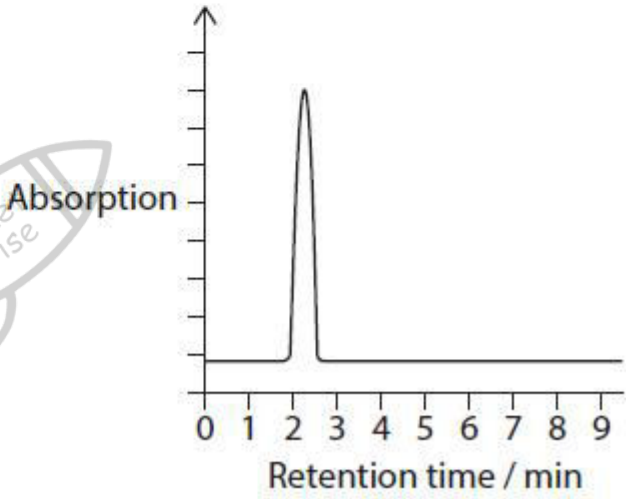


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☒ **B**



☒ **C**



☒ **D**

(Total for question = 3 marks)

Q37.

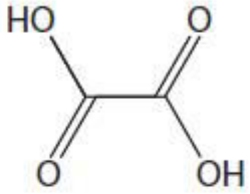
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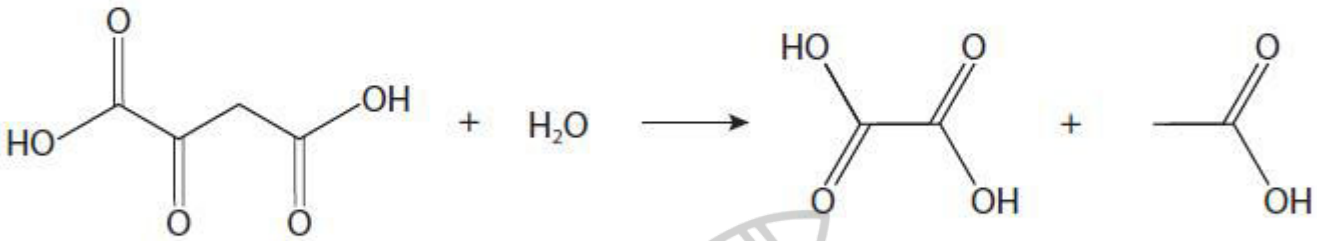
Oxalic acid

Oxalic acid has the IUPAC name ethanedioic acid and a molar mass = 90 g mol^{-1}.



The name oxalic acid originated from the fact that it was first isolated from plants of the genus *Oxalis*. The formation of oxalic acid in plants involves the hydrolysis of 2-oxobutanedioic acid.

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Plants can also form oxalic acid from glycolic acid, $HOCH_2COOH$.

Oxalic acid can be manufactured by oxidising sucrose using nitric acid in the presence of a vanadium(V) oxide catalyst. Over 100,000 tonnes of oxalic acid are produced each year.

Much of the oxalic acid produced is used for cleaning or bleaching, because it converts insoluble iron compounds into soluble iron salts.

(a) Calculate the atom economy, by mass, for the formation of oxalic acid from the hydrolysis of 2-oxobutanedioic acid.

(2)

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(b) Give the laboratory reagents and conditions required to convert glycolic acid, $HOCH_2COOH$, to oxalic acid.

Do not consider nitric acid.

(2)

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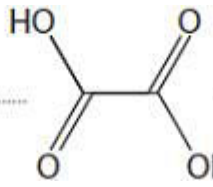
.....

(c) Sucrose is oxidised by concentrated nitric acid to form oxalic acid.

(i) Complete the equation for the reaction of one mole of sucrose with nitric acid.

(2)

$C_{12}H_{22}O_{11}$ + $36HNO_3$ ⟶ ..... + ... NO_2 + ... H_2O



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(ii) Give **one** observation that would be seen during this reaction.

(1)

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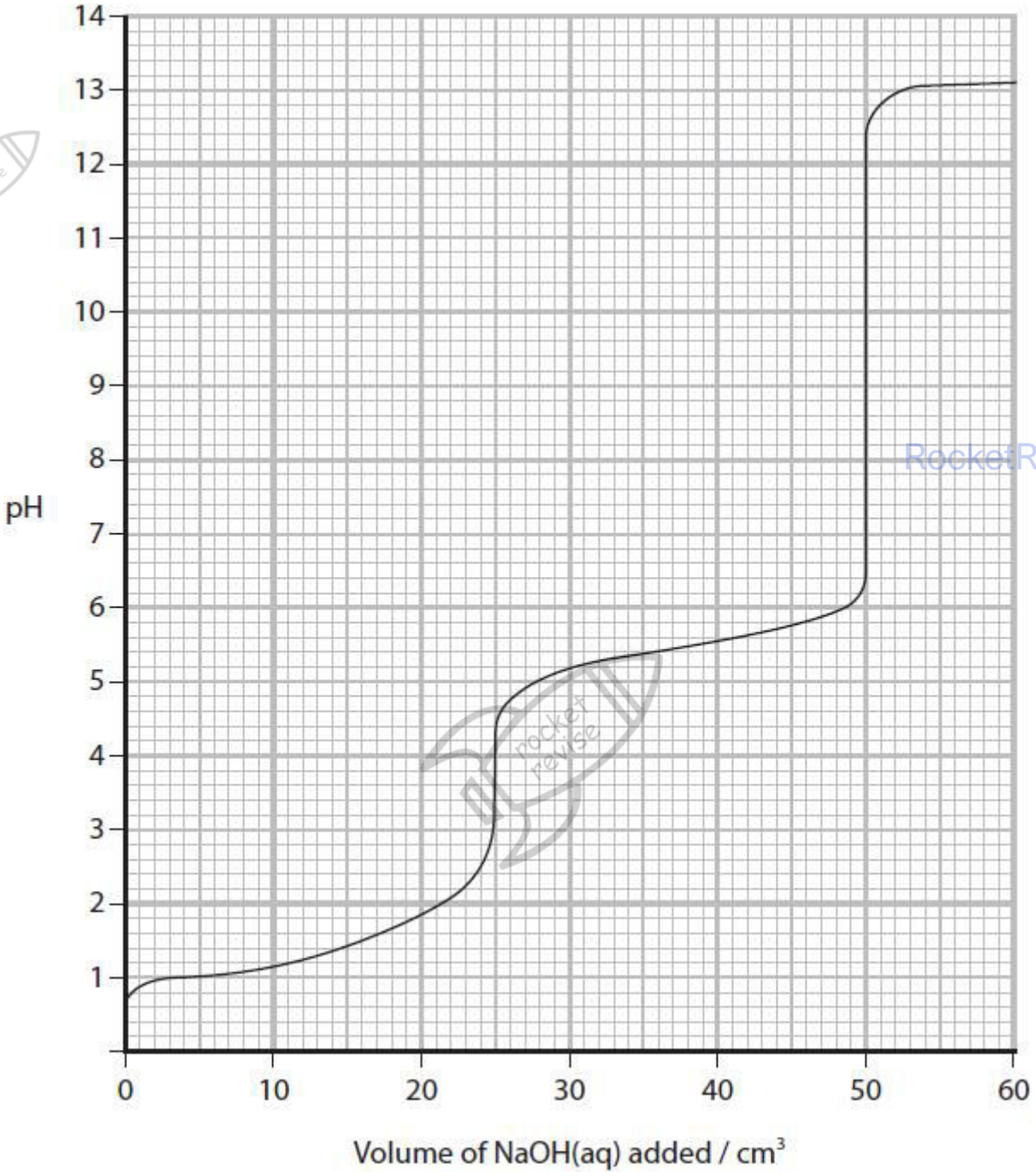
(d) A solution of sodium hydroxide was titrated against 25.0 cm^3 of oxalic acid solution. Both solutions were of equal concentrations.

(i) A student sketched a curve to show the changing pH during the titration.

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Select **one** indicator for each equivalence point from this titration graph. Include the colour at the end-point for each indicator. Use your Data Booklet.

(2)

First indicator

.....

Colour at the end-point

.....

Second indicator

.....

Colour at the end-point

.....

(ii) Suggest a difficulty that would occur if both indicators were present in the same mixture.

(1)

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.....

(iii) Determine, by using the graph, the **two** K_a values for oxalic acid.
You **must** show your working on the graph.

(5)

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(e) Spinach is one plant source of oxalic acid. Approximately 700 mg of oxalic acid can be extracted from 100 g of spinach.

(i) High-performance liquid chromatography, HPLC, is used for analysis in the extraction process. The HPLC equipment is connected to a detector which records peaks when substances pass through the HPLC column.
Suggest how a pure sample of oxalic acid would be used to identify the oxalic acid peak in the chromatogram of the spinach extract.

(1)

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.....

.....

(ii) A 0.558 mol dm^{-3} solution of oxalic acid is used for cleaning.
Calculate the mass of spinach required to make 500 cm^3 of this oxalic acid cleaning solution.
[Data: M_r oxalic acid = 90]

(3)

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(f) Write the equation for the reaction of oxalic acid with **excess** phosphorus(V) chloride.

(2)

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(Total for question = 21 marks)

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Q38.

*Proton NMR spectroscopy and mass spectrometry give detailed information about the structure of organic compounds.

Discuss the similarities and differences in these spectra for propanal and propanone.

Include in your answer:

Proton NMR Spectroscopy

- the number of peaks, the relative peak areas and, where relevant, splitting patterns

Mass spectrometry

- any *m*/*z* peaks that show one similarity and one difference between the spectra of the two compounds.

(6)

(Total for question = 6 marks)

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Q39.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

High resolution mass spectrometers can give peak data accurate to four decimal places.

An organic compound **X** has a molecular ion peak at *m*/*z* = 60.0210.

The accurate relative atomic masses are shown.

H = 1.0078, C = 12.0000, N = 14.0031, O = 15.9949

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Which compound is **X**?

- ☒ **A** $CH_3CH_2CH_2OH$
- ☒ **B** CH_3COOH
- ☒ **C** $CO(NH_2)_2$
- ☒ **D** $(CH_3)_3N$

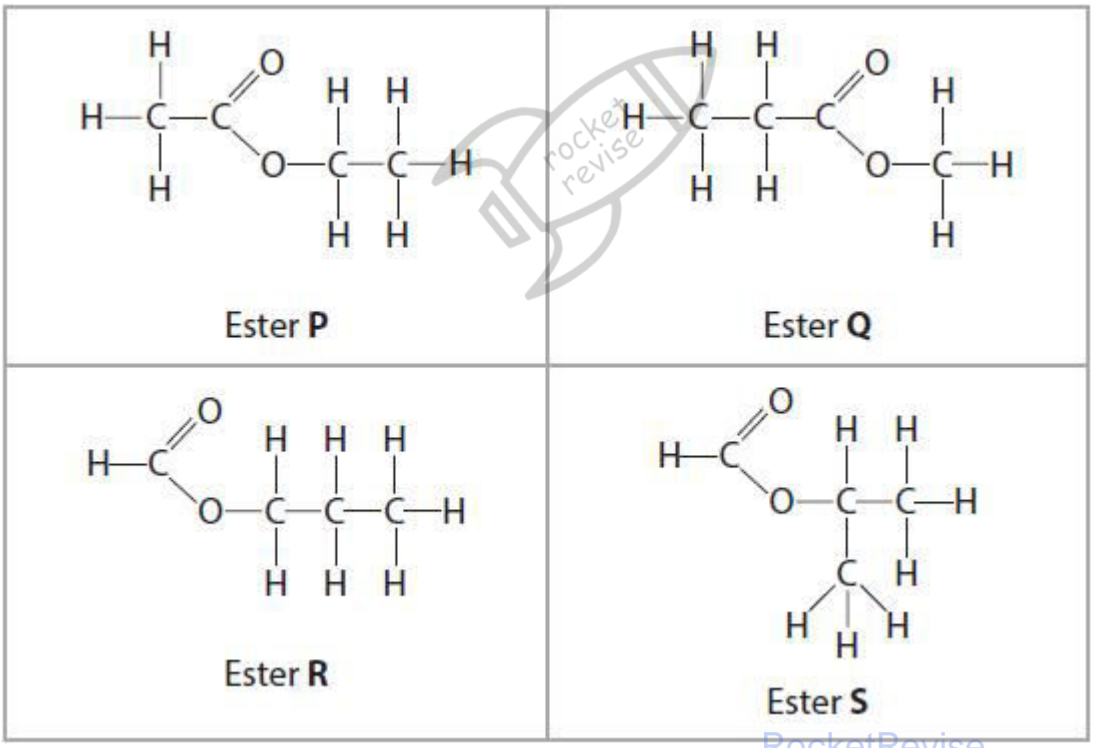
(Total for question = 1 mark)

Q40.

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Esters are difficult to distinguish from each other using chemical tests, so NMR spectroscopy is often used.

(a) There are four ester isomers with the molecular formula $C_4H_8O_2$. Their displayed formulae are shown.



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Explain which of the four esters can be distinguished from the other three by the numbers of peaks in the carbon-13 (^{13}C) NMR spectra.
Justify your answer by labelling any equivalent carbon atoms.

(3)

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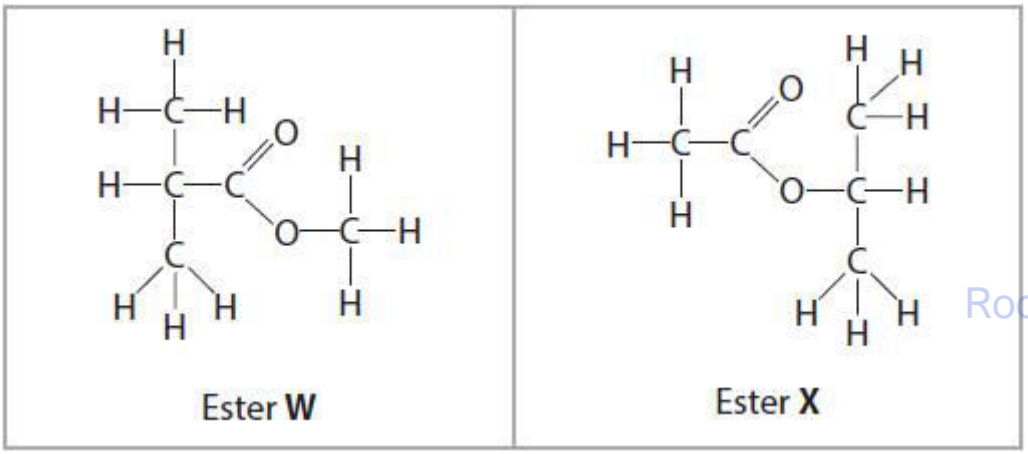
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(b) There are nine ester isomers with the molecular formula $C_5H_{10}O_2$.

(i) The displayed formulae of two of these esters are shown.



Ester **W**

Ester **X**

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Esters **W** and **X** both have one peak with a relative area of 3 in their **low**-resolution proton NMR spectra.
Explain how these peaks can be used to distinguish these two esters.

(2)

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(ii) The displayed formulae of two different esters are shown in the boxes.

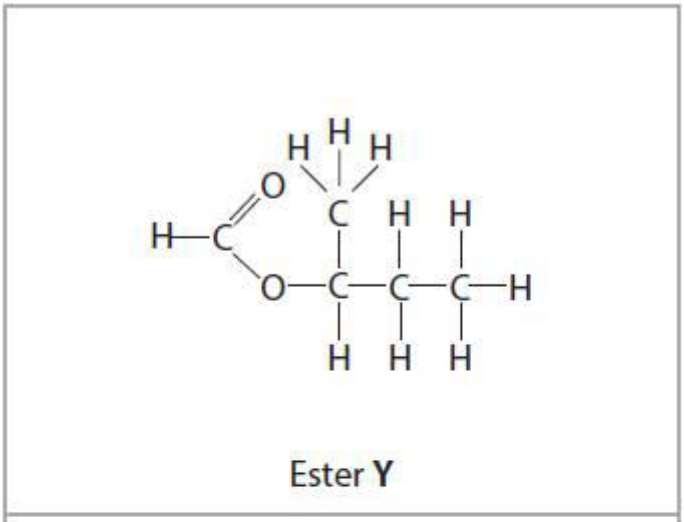
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Label the displayed formulae of the two esters **Y** and **Z** to show the proton environments and the splitting patterns of the peaks in the **high**-resolution proton NMR spectra.
Clearly indicate any equivalent proton environments.
For any multiplet, you may refer to the number of separate peaks in the multiplet rather than giving a name.

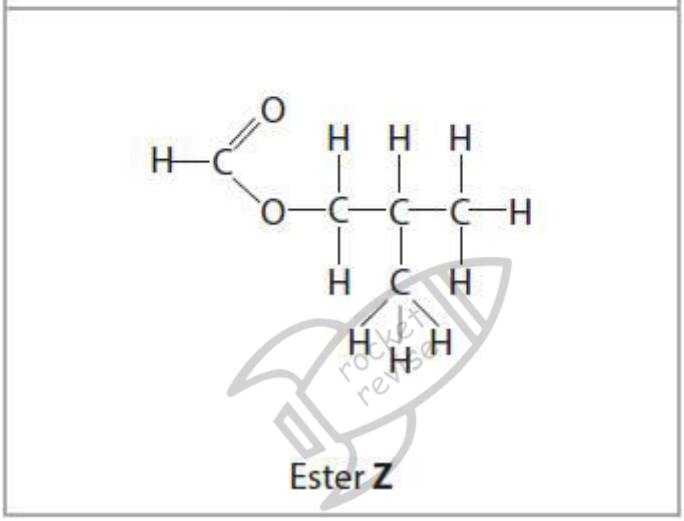
(4)

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Ester **Y**



Ester **Z**

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(iii) Draw the displayed formula of the $C_5H_{10}O_2$ ester which has **only** 2 peaks in its low-resolution proton NMR structure.

(1)

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(Total for question = 10 marks)

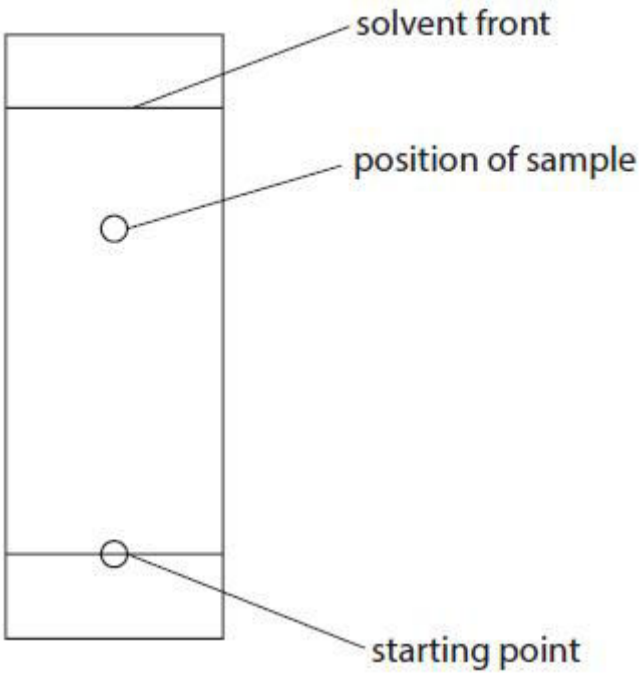
Q41.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A thin layer chromatogram of a substance is shown.

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What is the R_f value of the sample?

- ☒ **A** 0.27
- ☒ **B** 0.69
- ☒ **C** 0.73
- ☒ **D** 1.37

(Total for question = 1 mark)

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